

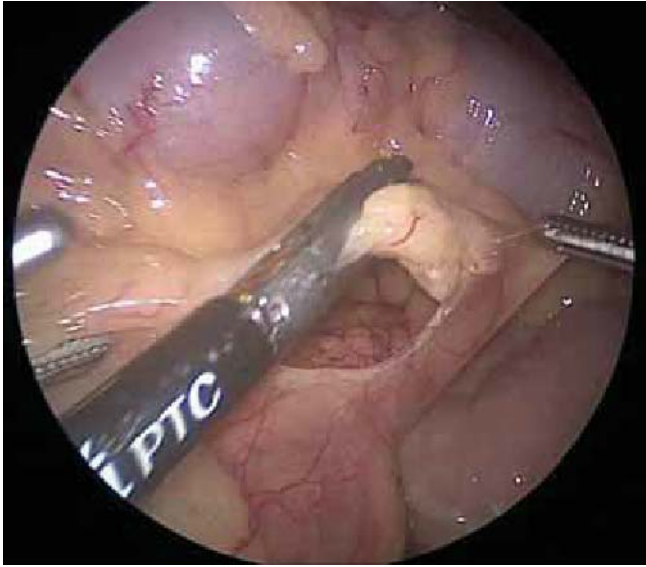


Figure 7.16 Creation of the mesocolic window to the left of the middle colic artery.

- The transverse colon is again elevated and a mesocolic window is created to the left of the middle colic artery (Fig. 7.16). The Roux limb is advanced through this window taking great care not to twist the mesentery and compromise the blood supply.
- We will describe two approaches to create gastrojejunal anastomosis. First the linear stapler technique will be described, and second, the circular stapler (end-to-end anastomosis [EEA]) technique will be outlined.
- Linear stapler technique:
 - Once the Roux limb is placed in the desired orientation, a running 2-0 silk posterior seromuscular suture line is placed between the Roux limb and the posterior wall of the distal stomach (Fig. 7.17).
 - Enterotomies in the stomach and small bowel are made to accommodate the 60 mm linear stapler with a blue load, and the side-to-side anastomosis is created (Fig. 7.18). The anterior layers of the stomach and jejunum are closed using a running 2-0 absorbable suture (Fig. 7.19).
 - The two-layer anastomosis is then completed anteriorly using a running 2-0 silk in the seromuscular layer (Fig. 7.20).

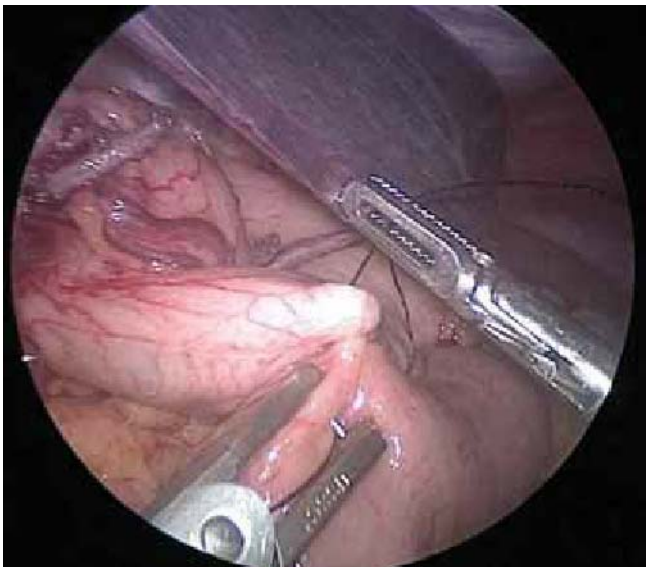


Figure 7.17 Once the Roux limb is placed in the desired orientation, a running 2-0 silk posterior seromuscular suture line is placed between the Roux limb and the posterior wall of the distal stomach.



Figure 7.18 Enterotomies in the stomach and small bowel are made to accommodate the 60 mm linear stapler with a blue load (3.5 mm staples), and the side-to-side gastrojejunal anastomosis is created.



Figure 7.19 The anterior layers of the stomach and jejunum are closed using a running 2-0 absorbable suture.



Figure 7.20 The two-layer anastomosis is then completed anteriorly using a running 2-0 silk in the seromuscular layer.

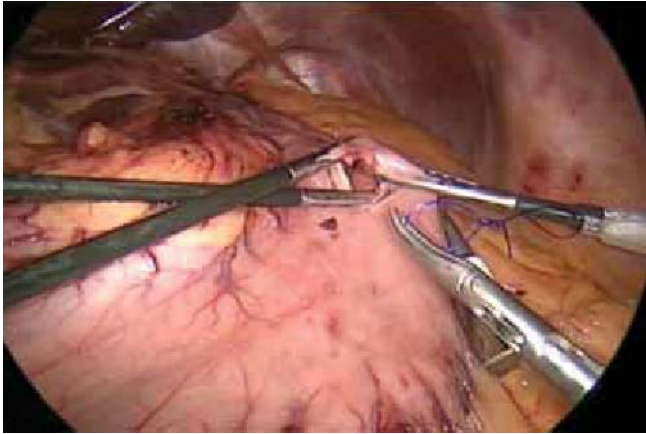


Figure 7.21 A distal anterior gastrotomy is performed to place the anvil inside the stomach. The anvil is then spiked through the posterior wall of the stomach in a location favorable for gastrojejunostomy. We use a 25 mm EEA stapler with 4.8 mm staples.

■ Circular stapler technique:

- This approach uses a 25 mm EEA stapler with 4.8 mm staples. A distal anterior gastrotomy is performed to place the anvil inside the stomach (Fig. 7.21). The anvil is then spiked through the posterior wall of the stomach in a location favorable for gastrojejunostomy.
- The anterior gastrotomy is then closed using either sutures or the linear stapler with 3.5 mm staples (Fig. 7.22).
- The Roux limb is then opened by removing the end staple line to accommodate the placement of the EEA stapler into the jejunum. The EEA stapler is advanced through the abdominal wall and inserted inside the Roux limb. The sharp post on the EEA stapler is delivered through the wall of the Roux limb 4 to 5 cm from the free margin. The stapler post and the anvil are mated, and the anastomosis is created by firing the stapler (Figs. 7.23–7.26).
- The redundant portion of the Roux limb is excised using a linear stapler with 2.5 mm staples (Fig. 7.27).
- It is our practice to perform an endoscopic air leak test after creation of gastrojejunostomy.
- The stomach is secured to the mesocolic window using interrupted 2-0 silk sutures to prevent herniation (Fig. 7.28).
- The specimen is retrieved and the wounds are closed using the surgeon's preferred method.

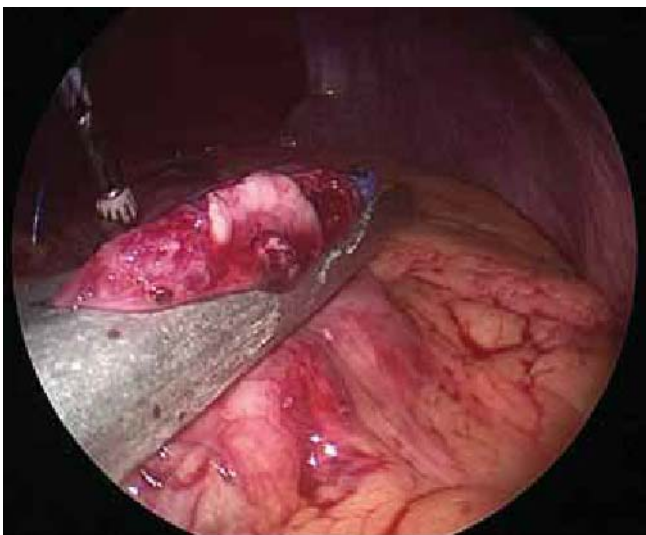


Figure 7.22 The anterior gastrotomy is then closed using either sutures or the linear stapler with 3.5 mm staples.

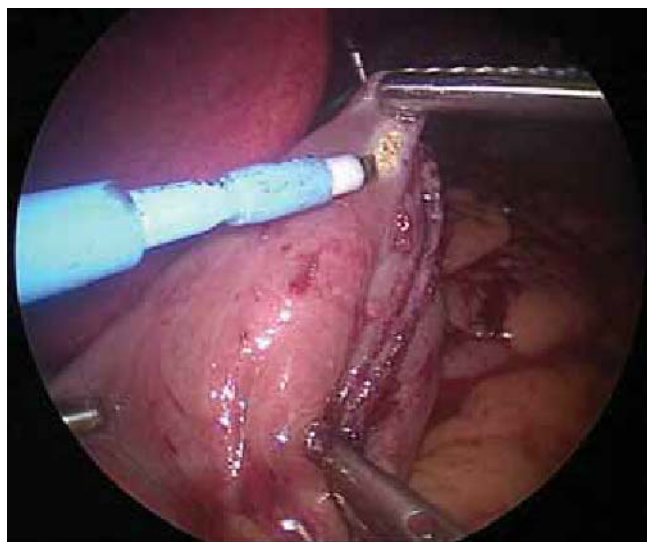


Figure 7.23 The Roux limb is then opened by removing the end staple line to accommodate the placement of the EEA stapler into the jejunum.

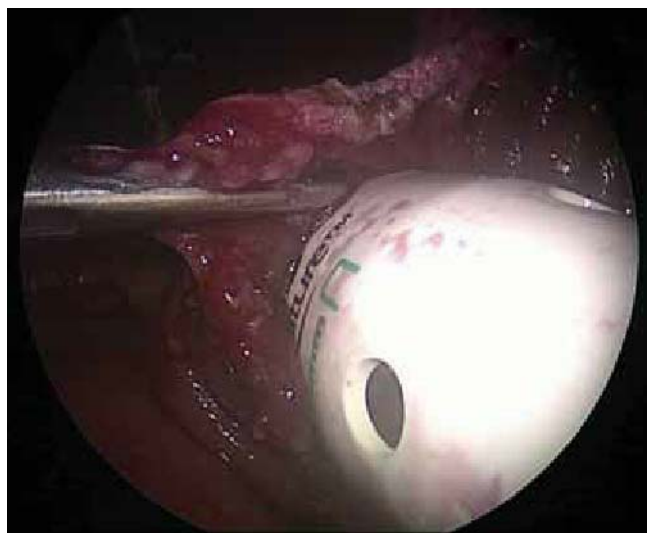


Figure 7.24 The EEA stapler is advanced through the abdominal wall and inserted inside the Roux limb.

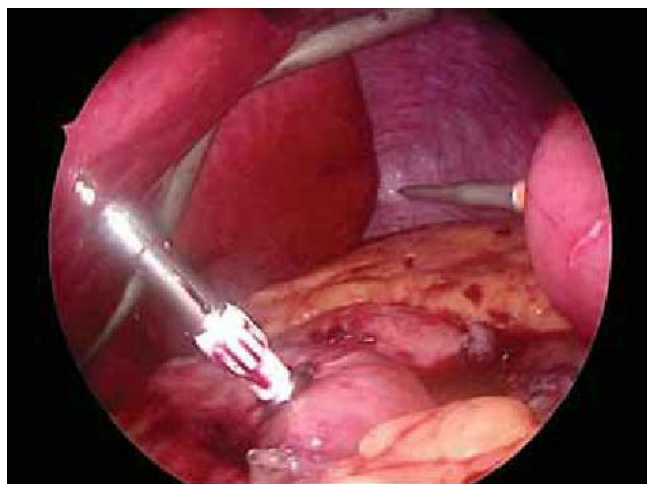


Figure 7.25 The sharp post on the EEA stapler is delivered through the wall of the Roux limb 4 to 5 cm from the free margin.

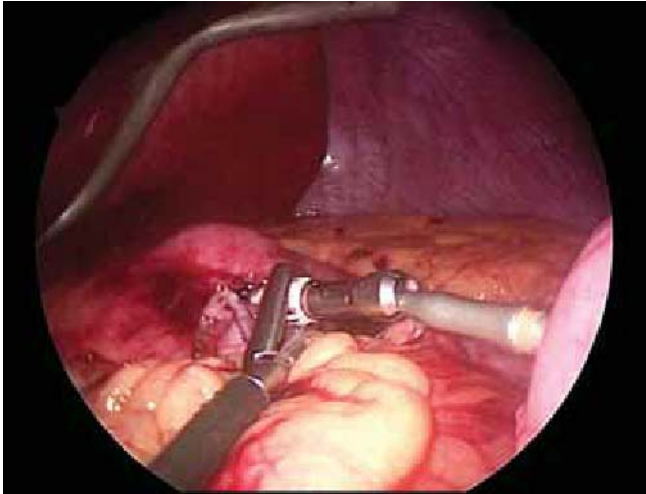


Figure 7.26 The stapler post and the anvil are mated and the anastomosis is created by firing the stapler.

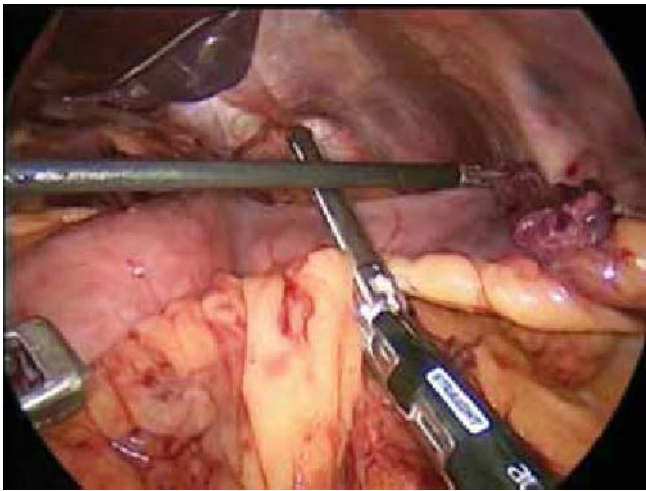


Figure 7.27 The redundant portion of the Roux limb is excised using a linear stapler with 2.5 mm staples.



Figure 7.28 The stomach is secured to the mesocolic window using interrupted 2-0 silk sutures to prevent herniation.



POSTOPERATIVE MANAGEMENT

The postoperative period for patients undergoing elective laparoscopic surgery for peptic ulcer disease is usually uneventful. On postoperative day 0, analgesia is provided with a patient-controlled anesthesia (PCA) pump. Ambulation and hourly incentive spirometer use are strictly enforced. The patient remains nothing per os (NPO), but nasogastric decompression is infrequently used.

On postoperative day 1 a liquid diet is introduced, IV fluids are discontinued and the urinary bladder catheter is removed. Nutritional counseling for a postgastrectomy diet is provided, and the patient is prepared for discharge on postoperative day 2.

Patients presenting with acute perforation or bleeding tend to have developed an inflammatory response and likely will require a longer hospital stay and closer monitoring.



COMPLICATIONS

Early complications after gastric surgery include bleeding, infection, unrecognized intra-abdominal injuries, and anastomotic leaks. The reported complication rates are quite variable but overall morbidity is 5% to 10% for the early complications (Table 7.1). Approximately 25% of the patients will develop symptoms associated with physiologic derangements after vagotomy and antrectomy. These disorders are collectively referred to as *postgastrectomy syndromes*.

- *Dumping syndrome* is an amalgamation of symptoms that occurs after rapid emptying of hyperosmolar chyme into the Roux limb, particularly carbohydrates. It exists in an *early phase* and a far less common *late phase*. The early phase of dumping usually occurs after 20 to 30 minutes following food ingestion, whereas the late form appears 2 to 3 hours after food consumption. Collective symptoms include epigastric distention, nausea, vomiting, palpitations, diaphoresis, dizziness, and flushing. In most patients mere avoidance of the offending nutrients results in symptom relief. In addition, frequent small meals low in carbohydrates is often helpful.
- *Postvagotomy diarrhea* is experienced by approximately 30% of the patients after truncal vagotomy. For the majority of these patients, the symptoms are self-limiting after 3 to 4 months and less than 1% will experience disabling diarrhea. Nonsurgical treatment including dietary and lifestyle modifications usually is sufficient for improvement and resolution. In more severe cases, antidiarrheals, octreotide, or cholestyramine to bind bile salts may be needed.
- *Delayed gastric emptying* or *gastric atony* is a multifactorial symptom complex that presents frequently after gastric resection with or without vagotomy. The most common

TABLE 7.1

Early Complications	Frequency (%)
Bleeding	1–10
Wound infection (open)	3–5
Pneumonia	5–10
Gastric stasis	4–5
Obstruction	1
Anastomotic leaks	1–4
Bowel perforation	1–3
Deep venous thrombosis	1–2
Cerebrovascular accident	<1

symptoms are early satiety, nausea, vomiting, and/or epigastric pain. Gastric atony can be diagnosed clinically, but radiologic, endoscopic, and scintigraphic studies can be used to confirm the clinical suspicion. The physician must consider other medical causes of gastroparesis, including diabetes mellitus, electrolyte imbalance, drug toxicity, and neuromuscular disorders. In addition, mechanical causes of gastric outlet obstruction such as anastomotic stricture, postoperative adhesions, Roux stasis, or internal herniation must be ruled out. As with postvagotomy diarrhea, the treatment is usually nonsurgical with diet and lifestyle modifications. Prokinetic agents such as metoclopramide or erythromycin can be used, but a significant clinical response is uncommon. In severe cases, a near total gastrectomy with Roux-en-Y reconstruction is an acceptable surgical option.

- *Roux stasis syndrome* has a similar clinical presentation to gastric atony. The etiology is not completely understood but is likely associated with alterations in jejunal motility. Construction of a Roux limb of more than 40 cm has been found to increase the incidence of Roux stasis syndrome and should be prevented when possible. Surgical treatment is reserved for those patients who do not respond to medical management and lifestyle changes. Reconstruction and shortening of the Roux limb to less than 40 cm is the surgical option of choice for these patients.
- *Recurrent ulcers* can be seen in approximately 1% to 2% of the patients undergoing truncal vagotomy and antrectomy. Incomplete transection or erroneously identifying the vagus nerve below its first hepatic or celiac branch can significantly jeopardize the outcome of the surgery. The rate of recurrent ulceration may be in direct proportion to the accuracy of identification and division of the vagus nerves. Additional technical factors that may lead to recurrent ulceration include retained gastric antrum and inadequate resection of the antrum.



RESULTS

Truncal vagotomy and antrectomy is infrequently performed due to improvements in medical management of ulcer disease, but when used for complicated cases it offers a 90% reduction in acid production. The mortality rate is approximately 1% to 2% with an overall complication rate of 5% to 10% (Table 7.2). With recent advances in surgical technology, the vast majority of these operations can now be performed laparoscopically; thus, decreasing surgical morbidity and improving patient satisfaction.



CONCLUSIONS

Truncal vagotomy with antrectomy is infrequently indicated in the treatment of peptic ulcer disease. However, this operation is reasonable for recurrent peptic ulcer disease. When possible, more physiologic operations such as Billroth I gastroduodenostomy or Billroth II gastrojejunostomy should be performed. Even though the indications for this operation have waned, the appropriate preoperative evaluation, technical performance, and postoperative management involved in this operation should be in the repertoire of all general surgeons.

TABLE 7.2

Truncal Vagotomy + Antrectomy	Results (%)
Morbidity	5–10
Mortality	1–2
Acid reduction	85–90
Ulcer reduction	1–2

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8 Proximal Gastric Vagotomy

Michael S. Nussbaum and Mark Alan Dobbertien

Introduction

Open proximal gastric vagotomy (PGV), as practiced now for over 40 years and known by other pseudonyms as highly selective vagotomy and parietal cell vagotomy, is a proven effective operation for the definitive surgical management of many complicated gastroduodenal ulcer disease patients and should be part of the armamentarium of all surgeons who care for such patients worldwide. This refined vagotomy operation has the basic surgical tenets of interrupting the vagal innervation to the parietal cell mass in the cardia, fundus, and body of the stomach (responsible for acid secretion) while maintaining the vagal innervation to the antrum and pylorus as well as to the branches of the celiac and hepatic plexus, thus minimizing motility and secretory disturbances in the stomach and the gastrointestinal tract. The effectiveness of PGV in lowering acid secretion has resulted in long-term cure of ulcer in over 90% of the patients and has had the added benefit of virtually eliminating the untoward side effects of less selective uses of vagotomy and drainage procedures or limited gastrectomy (antrectomy) such as dumping, stasis, bile reflux, and diarrhea. Although the PGV operation has been reported to be only performed once, on average, in training by graduating residents and is more technically demanding and time consuming (about 120 minutes average) than truncal vagotomy, the procedure can be performed safely and efficiently when done correctly and mastered. The many advantages of this procedure certainly outweigh the above disadvantages and they should not be considered a cogent rationale to abandon the procedure when performing definitive ulcer surgery.

In this chapter, the indications for the operation will be outlined and the preoperative, operative, and postoperative management for patients undergoing PGV will be presented along with the expected results and complications associated with open PGV.



INDICATIONS

PGV is indicated in patients with chronic duodenal ulcers who have failed the currently available maximal medical management regime. It is also sometimes the procedure of choice in stable patients who are not ulcer naïve and have a perforated or bleeding duodenal ulcer and are able to tolerate a definitive operative ulcer procedure once a Graham patch (or accepted alternative) or artery ligation has been performed. In addition,

PGV can be used in active ulcer patients receiving medical management who need aortic surgery or organ transplantation to prevent serious, life-threatening ulcer complications such as bleeding or perforation. PGV alone should not be used in patients with pyloric or prepyloric ulcers or for gastric outlet obstruction as the recurrence rate and failure rate are 31% and 55%, respectively, unless a drainage procedure is added to PGV. The indication for PGV in type I gastric ulcer in addition to wedge resection or submucosal resection remains controversial.



PREOPERATIVE PLANNING

Proper preoperative evaluation and management of patients necessitating either elective or emergency surgery using PGV should not be underestimated. Any lack of attention to detail may result in untoward morbidity and mortality. In all cases, a detailed history and physical examination is required as is a review of up-to-date laboratory and imaging data. A preoperative assessment of the patient's cardiopulmonary risk factors, ASA PA class, APACHE score as well as entering data in the American College of Surgeons National Surgical Quality Improvement Program (ACS NSQIP) data base (if available) may assist in the proper management of patients needing elective and emergency PGV. Of particular importance is a detailed review of the individual patient's ulcer diathesis history, endoscopic findings, pathology, and treatment up to that point. The emergency surgical management of duodenal ulcer complications, such as perforation and bleeding, is often predicated on the previous history of ulcer care prior to their presentation. After the above review and assessment, therapies and technical considerations are manipulated to minimize cardiopulmonary morbidity and mortality, venous thromboembolism (VTE), and surgical site infection. In order to minimize perioperative myocardial infarction, patients with intermediate to high-risk scores are treated with beta blockers without intrinsic sympathomimetic activity (unless contraindicated), and this has resulted in a greater than 50% relative reduction in postoperative cardiac events. High-risk pulmonary patients would benefit from incentive spirometry, selective nasogastric tube use, and short-term neuroaxis blockage via epidural anesthetic techniques. Evidence shows that patient-related risk factors, such as chronic obstructive pulmonary disease, age older than 60 years, ASA class of II or higher, functional dependence, congestive heart failure, prolonged surgery, and abdominal surgery, increase the risk for postoperative pulmonary complications. All patients should receive VTE prophylaxis with either unfractionated or low-molecular weight heparin plus sequential compression devices may be used in addition to anticoagulation in the high-risk surgical patient and in those who have a contraindication to anticoagulation prophylaxis. Caution is advised in using low-molecular weight heparin in patients with renal insufficiency or failure, and it either must be dose-adjusted or unfractionated heparin must be used as a substitute. For most cases of uncomplicated gastroduodenal ulcer surgery, no prophylactic antibiotics are recommended. However, in complicated patients or patients where the gastrointestinal tract will be opened, a first-generation cephalosporin (i.e., cefazolin) should be administered within 1 hour of the surgical incision to reduce the likelihood of a postoperative surgical site infection.



SURGICAL TECHNIQUES

Positioning, Anesthesia, and Skin Preparation

The patient is positioned on the operating table in the supine position. At least one arm should be tucked in most instances to provide more room for the operating team and any mechanical retractors. Care must be taken to ensure that the patient is properly off-loaded to prevent nerve injury in both the upper and lower extremity and to prevent back pain in the lordotic areas of the lumbar and cervical spine. A footboard should be

used so that the patient will not move during the procedure when reverse Trendelenburg positioning may cause shifting of the patient on the operating table. Patients should be secured to the operating table using a safety strap. The patient should have a large bore intravenous line and, in the high-risk patient, may need central venous pressure monitoring as well as invasive blood pressure monitoring with an arterial line. A urinary catheter is placed for urine output monitoring and a nasogastric tube is placed once the patient has undergone rapid sequence intubation and the endotracheal tube has been secured. The nasogastric tube is helpful for decompression of the stomach as well as for identifying the esophagus at the diaphragmatic hiatus and as a retraction device on the greater curve of the stomach. Patients should be offered perioperative thoracic epidural analgesia which should be placed prior to the induction of general anesthesia. All patients undergoing PGV should be considered to have a full stomach and therefore require rapid sequence intubation techniques. Muscle relaxation is necessary for open PGV. The patient's abdomen and lower chest is clipped in the operating room and the patient's skin is prepared with chlorhexidine-alcohol, which has recently been shown to be superior to povidone-iodine in preventing postoperative surgical site infection. Sterile draping should include the lower chest well above the xiphisternum.

Incision, Exposure, Exploration, and Definitive Management of Bleeding and Perforation

An upper midline, epigastric incision is recommended when performing open PGV and provides excellent overall exposure. Infrequently, the incision needs to be carried below the umbilicus. An alternative chevron incision may be used, but the disadvantages of muscle transection seem to outweigh the exposure benefits in most cases. Care must be taken to avoid injuring the periosteum of the xiphoid process in order to prevent the rare but under-reported incidence of heterotopic ossification of the abdominal wound. The xiphoid, in most instances, does not need to be excised. Once the skin has been incised, blunt bilateral force is used to transect the subcutaneous tissue in a bloodless field, and this technique reliably identifies the midline fascia and linea alba in situations when previous laparotomies have not been performed. The midline fascia is incised precisely in the midline to avoid disruption of the confluence of the anterior and posterior rectus sheaths. The peritoneal fat is sharply transected and the peritoneum is opened to the full extent of the wound, avoiding any adhesions when present. The ligamentum teres hepatis does not need to be transected in most cases and may be used to patch a perforated duodenal ulcer when omentum is not readily available. Full abdominal exploration is mandatory in order to assess the suspected pathology as well as identify any other unsuspected pathology, unless mitigating circumstances would call for a truncated exploration, such as dense adhesions, when the risk of enterotomy may outweigh the benefits of exploration. Any peritoneal soiling is quickly addressed with suction, irrigation, and control of ongoing spillage from the perforated viscous. The left triangular ligament may be incised to provide right upward mobilization of the lateral segment of the left lobe of the liver for better exposure of the esophageal hiatus. A laparotomy sponge is placed lateral and posterior to the spleen to release any tension on the gastrosplenic ligament, thus minimizing the risk of splenic injury. A self-retaining retractor is next implemented and attached to the operating table. A Thompson or Bookwalter retractor is preferred over a Balfour retractor in most cases. In order to provide the best exposure and retraction, the liver needs to be retracted superiorly and to the right and the parietes need to be placed on tension inferiorly and mediolaterally. The upper abdominal retractors should be placed with the patient in Trendelenburg position. Once the retractors are in place, reversing the Trendelenburg position will provide optimal exposure of the upper abdomen. All of the many important surgical aspects of PGV and the emergency management of perforation and bleeding will be able to be addressed without any further manipulation of the retractor system. In clinical scenarios of duodenal ulcer hemorrhage, an anterior longitudinal gastroduodenostomy across the pylorus is performed and the bleeding artery is ligated, making sure to control the transverse pancreatic artery with a U stitch. After controlling the arterial hemorrhage

the duodenotomy is closed horizontally in a Heineke–Mikulicz fashion. A perforated duodenal ulcer should be patched with well vascularized omentum when available and should not be closed primarily especially if the closure would create narrowing of the duodenum. Closure and patching is an acceptable modification of Graham's technique, when the ulcer is small and would not result in significant stricturing. The PGV can be added to both of these procedures when indicated. A chronically scarred and obstructed duodenum should be managed either by resection or by PGV with gastrojejunostomy to alleviate the obstruction and to minimize recurrent duodenal ulcer.

Proximal Gastric Vagotomy

PGV requires a surgeon with a tremendous requisite knowledge of vagal and foregut anatomy and a meticulous technical skill-set for best results. The procedure should be able to be performed in less than 2 hours and consists of four (or five) important steps for optimal denervation (completeness of vagotomy), preservation of antral-pyloric mobility, and prevention of ulcer recurrence. The procedural steps involved in PGV are that of anterior lesser omental dissection, posterior lesser omental dissection, distal esophageal dissection, lesser curvature closure, and possibly consideration of extended PGV in the area along the greater curvature across the incisura described by Rosati in order to bring cure rates as high as 98%. The steps in the operation are not necessarily done sequentially and may be altered depending on local conditions for a given patient.

Distal Esophageal Dissection

The PGV is usually begun at the level of the esophageal hiatus and the distal esophagus (Fig. 8.1). The distal esophageal dissection and proximal lesser curvature dissection along the cardia and fundus can be very challenging to surgeons who are unfamiliar with the anatomy of the right and left vagus nerve as they descend in the posterior mediastinum (right) and through the esophageal hiatus. The dissection plane is not limited in scope and should be envisioned as a 360-degree operating plane around the distal 5 to 10 cm of esophagus and the proximal dome of the stomach's cardia and fundus. The left and right vagus nerves form a plexus over the posterior mediastinal esophagus and eventually unite to form the trunks of the anterior and posterior vagus at the level of the distal esophagus before dividing into divisions at or near the esophageal hiatus, but certainly above the angle of His. For this reason, it is recommended to begin this portion of the procedure at the angle of His by encircling the esophagus at

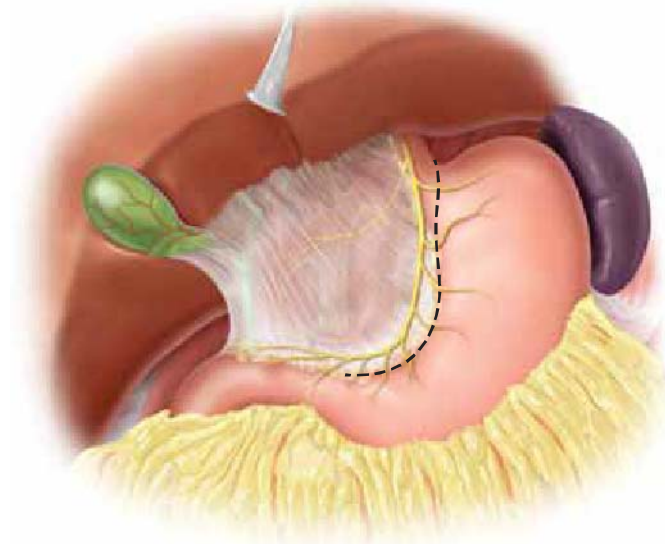


Figure 8.1 Anterior view of the stomach demonstrating the anterior nerve of Latarjet. The dotted line represents the line of dissection. The dissection is begun at the esophagogastric junction and the vagus nerves are isolated prior to the lesser curvature dissection.

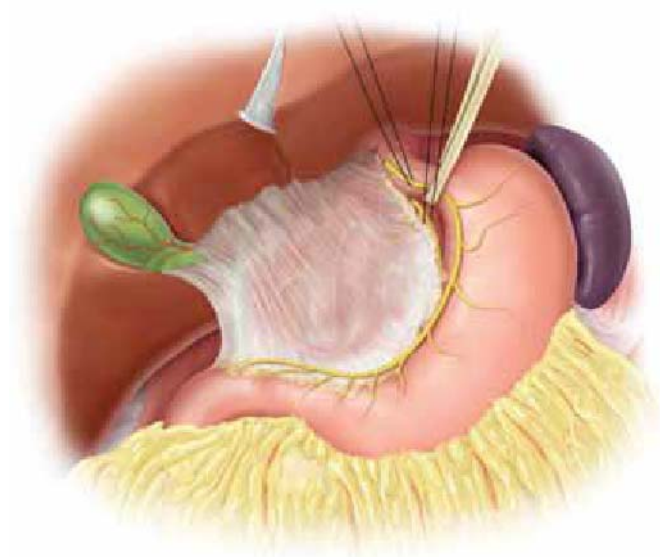


Figure 8.2 The lower esophagus is exposed and the gastroesophageal junction is encircled with a Penrose drain and retracted inferolaterally. The anterior and posterior nerves of Latarjet are secured with stays for proper identification, preservation, and retraction.

the cardioesophageal junction with a Penrose drain so that downward traction can be applied on the junction tenting the nerves for easy identification. In addition, starting the dissection at this point provides the best window for free mobilization of the esophagus and proximal stomach from the vagi, which are usually not as intimate with the esophageal musculature at this anatomical area. The encircling of the esophagus at the angle of His is accomplished by incising the peritoneum in this area and at the phrenicoesophageal ligament overlying the right crus of the diaphragm passing a blunt instrument from left to right, thus encircling the esophagus at the cardioesophageal junction (Fig. 8.2). Next, the anterior and posterior vagal trunks are encircled with umbilical tapes or vessel loops. The anterior nerve is usually intimately draped obliquely across the distal anterior esophagus and should be mobilized to the right of the esophagus. Occasionally, two divisions are seen before forming the anterior nerve of Latarjet. The hepatic branch can be seen in the lesser omentum running toward the caudate lobe of the liver. The larger posterior nerve is usually not intimate with the muscular wall of the esophagus as it enters the abdomen and is located typically at 8 o'clock on the esophagus close to the right crus of the diaphragm. The celiac nerve, a division of the posterior vagus, runs in the lesser omentum toward the left gastric artery, a branch of the celiac trunk. The phrenicoesophageal ligament is then incised completely and the distal esophagus is mobilized proximally into the mediastinum for approximately 5 to 7 cm to ensure complete esophagogastric vagotomy.

Anterior Lesser Omental Dissection

Once the vagal trunks are isolated and separated from the esophagogastric region, attention should be directed to the anterior lesser omentum. In order to provide distraction between the lesser omental contents and the lesser curvature of the stomach, an incision is made in the pars flaccida to provide superior retraction on the lesser omentum. This technique also provides a window into the upper recess of the lesser sac and exposure to the right crus of the diaphragm where the posterior vagus nerve is usually identified posterior and medial to the esophagus. The stomach is retracted inferiorly, either with direct anterior greater curvature retraction or by incising the gastrocolic omentum distal to the gastroepiploic vessels and dividing some anterior epiploic vessels such that combined anterior and posterior gastric wall inferior retraction can be applied. The anterior nerve of Latarjet is identified 1 to 2 cm from the lesser curve and must always be visualized directly during the procedure to avoid injury. Distraction of the lesser omentum helps prevent injury to the nerve. Location of the anterior nerve of Latarjet's oblique ending in the crow's foot with its three typical end branches is another important landmark

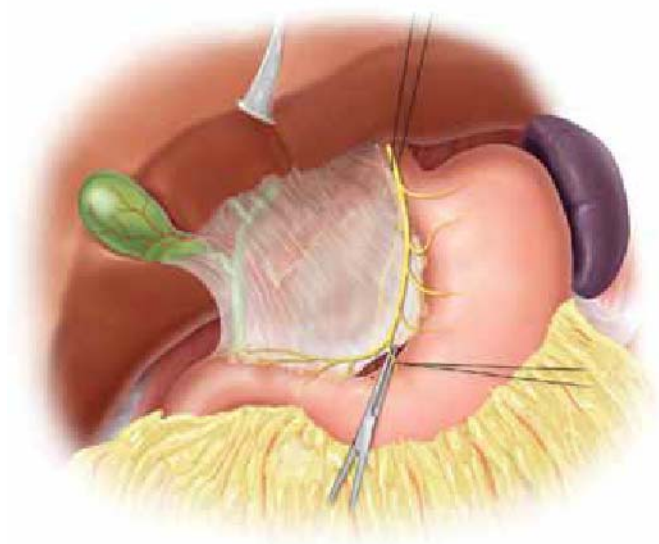


Figure 8.3 The anterior lesser curvature dissection is begun 5 to 7 cm proximal to the pylorus, and each gastric vagal branch from the anterior nerve of Latarjet and their accompanying vessels are isolated, ligated, and divided. Occasionally, when the crow's foot is proximal on the antrum, the heel of the crow's foot may be used as a starting point.

as is the 5- to 7-cm mark proximal to the pylorus on the lesser curvature (Fig. 8.3). In some patients the crow's foot is relatively proximal on the antrum and may not represent the best starting position to begin the lesser curvature devascularization and end gastric branch vagotomy. In such situations, either the heel of the crow's foot (most proximal neurovascular bundle of the crow's foot) should be transected in order to achieve complete parietal cell vagotomy without compromising antral motility. In general, the anterior lesser omental dissection begins about 5 to 7 cm from the pylorus. The peritoneum overlying the anterior leaf of the lesser omentum is then scored sharply from the distal dissection margin at the antral-body junction as close to the serosa of the anterior lesser curvature of the stomach as possible without injuring the stomach. The peritoneum is scored all the way from the incisura along the lesser curve across the cardioesophageal junction below the esophageal fat pad and across the angle of His. For surgeons who routinely incise the gastrocolic ligament, the posterior peritoneum corresponding to the above mentioned anterior peritoneum may be scored as well once all gastropancreatic adhesions are cleared and the stomach is retracted anteriorly and superiorly. Certainly, releasing both sleeves of the peritoneum will increase the distance between the nerves of Latarjet and the lesser curvature of the stomach reducing the likelihood of nerve injury and converting a parietal cell vagotomy to a total gastric vagotomy, defeating the benefits of the procedure and requiring a drainage procedure for salvage. Care must be taken in not applying too much tension on the lesser omental structures as this could cause avulsion of the end gastric branches of the ascending and descending left gastric artery. Once the peritoneum has been scored, the surgeon should be cognizant of the concave and, reciprocally, convex nature of the relationships between the lesser curvature of the stomach and the perpendicular neurovascular end gastric branches. This important relationship can be manipulated either by proper movement of surgical ligation and division devices or by rolling the lesser curvature of the stomach more anteriorly and inferiorly as the lesser omental dissection proceeds more posteriorly. In either case, ligation and division of the neurovascular bundles should be done as close to the stomach as possible without injuring the gastric wall. Both techniques can be used in tandem during the lesser curvature devascularization and vagotomy if the surgeon chooses so. It is best to divide the perpendicular neurovascular bundles as they enter the lesser curvature with a 2-0 or 3-0 silk suture using a fine right angle clamp and tying the pedicles in continuity, limiting the possibility of injury to the surrounding structures and blood loss from inappropriate tension on these fine neurovascular bundles that clamps may precipitate. An alternative approach is to use an energy source to provide ligation and division. The ultrasonic shears, Ligasure, or Enseal devices may be used to facilitate this dissection, and all these devices have been used safely and

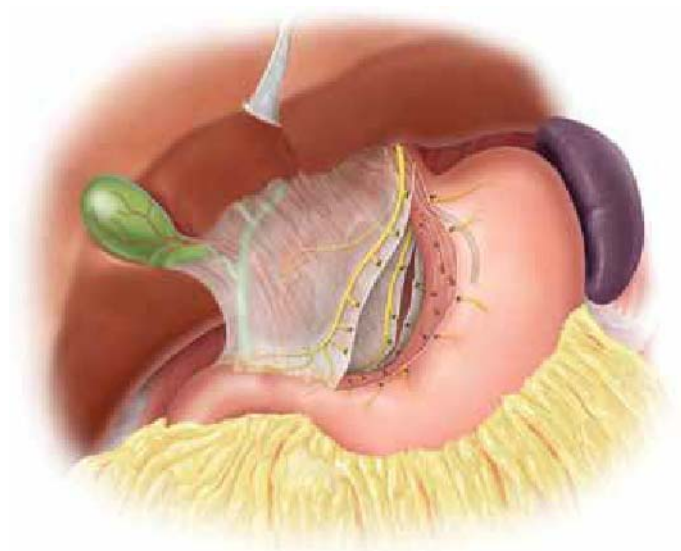


Figure 8.4 The lesser curvature dissection consists of three layers of nerves and vessels—an anterior, posterior, and unpredictable middle layer—all of which must be divided to ensure a complete proximal gastric vagotomy. The nasogastric tube is important for both gastric decompression and inferior gastric retraction.

effectively in minimally invasive vagotomy procedures. Either way, care must always be taken to avoid injury to the anterior nerve of Laterjet and the lesser curvature of the stomach. The anterior gastric nerve transection should continue to the prior proximal dissection at the cardioesophageal junction to ensure that the anterior and posterior nerves of Laterjet and the hepatic and celiac branches are identified before beginning the posterior lesser omental dissection. Care must be taken not to apply too much tension on the vagal trunks which could produce neuropraxia, or worse, traction neurolysis.

Posterior Lesser Omental Dissection

Before beginning the posterior lesser omental dissection, it is important to realize that there are numerous and variable small gastric vagal branches from both the anterior and posterior nerves of Laterjet that innervate the stomach's lesser curve and arise in the midportion of the gastrohepatic sleeve of mesentery (Fig. 8.4). It is imperative that these branches be ligated and divided with precision as most of the times they are not accompanied by a gastric end artery. This middle sleeve of mesentery should be ligated and divided in continuity and carried out from anterior to posterior and from inferior to superior extending up to the cardioesophageal junction. This procedure is best performed close to the lesser curve of the stomach and can be facilitated by rolling the lesser curve anteriorly and inferiorly. At this point, the posterior sleeve of the mesentery with its posterior nerve of Laterjet and the gastric branches of the descending branch of the left gastric artery to the posterior wall of the lesser curvature of the stomach should be easily identified. Dissecting the midportion of the mesentery en bloc with the posterior lesser omental dissection is not recommended. In obese patients, the midportion of the mesentery should be dissected in multiple layers to ensure completeness of the vagotomy. The posterior lesser omental dissection should be performed as described for the anterior lesser omental dissection. Finger dissection and pinching of the posterior lesser omentum is occasionally helpful and can disperse the omental adipose tissue from the surrounding neurovascular pedicles ensuring that the posterior nerve of Laterjet is not injured. The dissection is carried out from distal to proximal on the posterior lesser curvature up to the cardioesophageal junction dissection. The two sleeves of mesentery should be inspected to ensure that the anterior and posterior nerves of Laterjet have been preserved and inserted adequately on the anterior and posterior aspect of the antrum 5 to 6 cm from the pylorus. The distal margin of neuro/devascularization on the antrum should be re-inspected for adequacy of vagotomy and all remaining vagus nerve tissue in this area should be divided (Fig. 8.5).

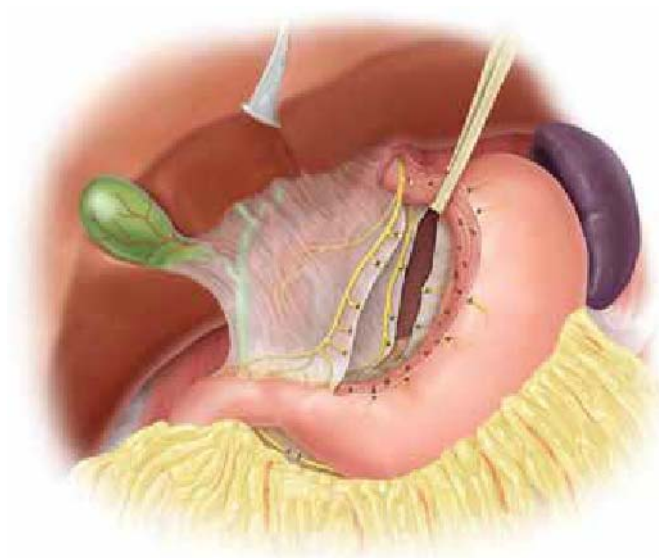


Figure 8.5 The completed proximal gastric vagotomy. All vagus fibers to the distal 5 to 10 cm of the esophagus are divided sharply making sure to divide the criminal nerve of Grassi in the gastropancreatic fold.

Finally, with minimal traction on the anterior nerve of Latarjet and inferolateral traction on the cardioesophageal junction, all structures to the left of the vagus entering the esophagus or stomach are ligated. Next, the same traction is applied to the posterior nerve of Latarjet and all the neurovascular structures to the left of the nerve are transected. The criminal nerve of Grassi, a branch of the posterior vagus, is identified and transected in the gastropancreatic fold as it makes its way to the fundus posteriorly. The esophagus is re-inspected ensuring that no gastric branches are imbedded in the esophageal musculature and no piercing mucosa are left untransected. Gentle blunt dissection with a finger or sharp dissection may be used to complete the esophageal portion of the vagotomy.

Lesser Curvature Closure

The open mesenteric surface of the lesser curvature of the stomach is closed by placing Lembert sutures using interrupted 2-0 or 3-0 silk (Figs. 8.6 and 8.7). This technique may prevent neural regeneration of the vagal fibers, potentially implicated in some incomplete vagotomies. In addition, this seromuscular closure of the lesser curvature of the stomach may avoid the potential complications of ischemic necrosis or perforation of

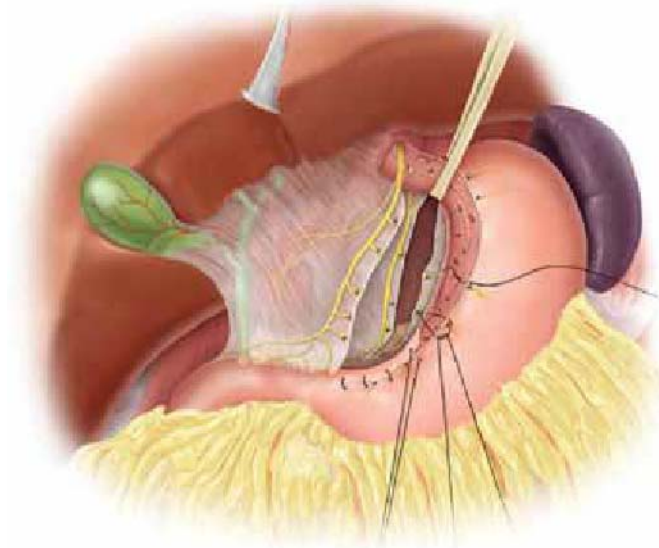


Figure 8.6 Lesser curvature closure is accomplished by approximation of the anterior and posterior seromusculature layers over the area of lesser curvature bare muscle in order to prevent free perforation from ischemic necrosis or from salient full thickness injury.

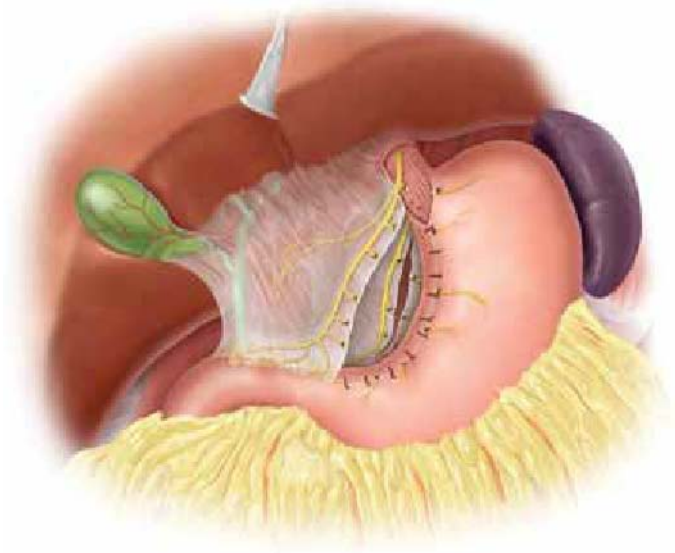


Figure 8.7 The completed seromuscular closure of the lesser curvature.

the lesser curvature of the stomach. Although these complications are exceedingly rare, a greater than 70% mortality rate has been reported in some series that experienced lesser curvature perforation as a complication. The likelihood of lesser curvature ischemic necrosis would be extremely rare, given the extensive submucosal blood supply, and therefore, these perforations are likely a result of unrecognized iatrogenic injury to the gastric wall during dissection. Crural repair can be carried out at this point when hiatal hernia is present, and fundoplication may be indicated when significant gastroesophageal reflux is documented preoperatively.

Extended PGV

When ulcer recurrence would be an untenable consequence for certain patients undergoing PGV, an extended PGV may be performed and has been reported to decrease ulcer recurrence rates to 2% versus 10% reported in most series. The “seven areas of vagotomy” described by Nyhus and his colleagues at the University of Illinois may serve as an anatomical template for surgeons performing PGV and extended PGV and may reduce the incidence of incomplete vagotomy (Fig. 8.8). Rosati et al. have recommended

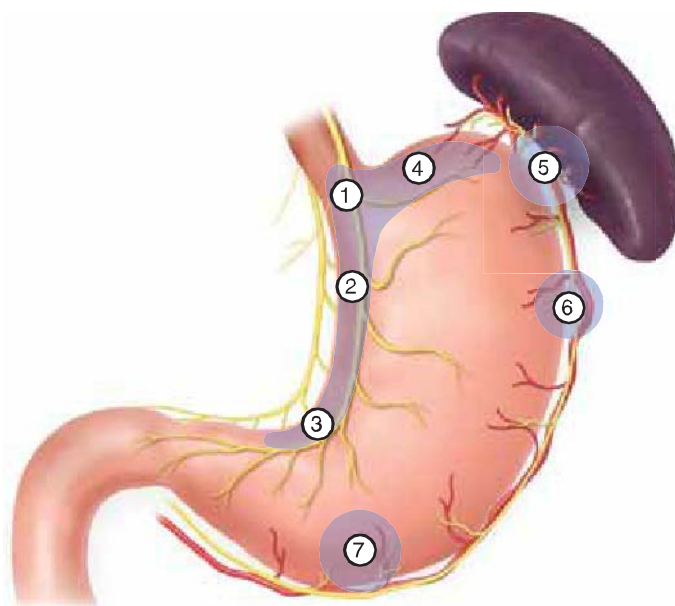


Figure 8.8 The seven areas of vagotomy are (1) periesophageal, (2) lesser curve, (3) crow's foot, (4) gastropancreatic fold, (5) short gastric, (6) left gastroepiploic, and (7) right gastroepiploic. Areas 3, 4, 6, and 7 are routinely divided during extended PGV while area 5 is always preserved to preserve blood supply to the proximal stomach.

that the preganglionic vagal fibers along the greater curvature across the incisura of the stomach (area 7) be divided to provide a more complete vagotomy. This additional procedure can be done with impunity for patients requiring PGV especially with regard to gastric blood supply and potential ischemic necrosis by avoiding disrupting the main right gastroepiploic vessels located near the gastric wall. The nerve fibers and end vascular gastric branches should be ligated and divided in continuity, avoiding injury to the gastroepiploic vessels. Energy devices for ligation and division may be used as an alternative to ligation in-continuity techniques.

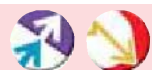
Abdominal Closure

The abdomen is closed once the PGV is completed and the instrument, sponge, and needle counts are correct. Copious warm isotonic saline or antibiotic lavage can be performed and the effluent should be grossly clear. All surgical sites are inspected for hemostasis and potential injury to the nerves of Laterjet or the abdominal hollow or solid viscera. All viscera are then placed back in their anatomical locations and the omentum is preserved anteriorly. A film of hyaluronic acid and carboxymethylcellulose may be added prior to fascial closure in an attempt to decrease adhesion formation to the anterior abdominal wall as well as to decrease the complexity of potential reoperative surgery. Closure of the midline fascia should be done with a running monofilament suture. The skin can either be stapled or closed with a running subcuticular absorbable suture.



POSTOPERATIVE MANAGEMENT

Postoperatively, the routine use of a nasogastric tube is not necessary and may cause morbidity in patients with underlying pulmonary disorders. VTE prophylaxis should be continued throughout the hospitalization unless contraindicated. Pain control should be provided via the epidural catheter that was placed prior to induction, either continuously or via patient-controlled measures. If epidural analgesia is not used then a patient-controlled analgesia device should be used. Early feeding is recommended starting with clear liquids and subsequently with a postgastrectomy diet. Acid reduction therapy is not required after PGV unless it is an integral part of therapy for *Helicobacter pylori* eradication. Resumption of all preoperative medications indicated for significant medical comorbidities should be started as soon as possible. Good pulmonary toilet and early ambulation is recommended. Close clinical observation for the development of complications is mandatory. Patients are typically discharged within 3 to 5 days after an uncomplicated procedure.



EXPECTED RESULTS AND COMPLICATIONS

Open PGV will yield excellent results with few complications for ulcer patients when the procedure is performed for the proper indications by a surgeon with the requisite skill and knowledge outlined above. The attractive features of PGV are its low morbidity and mortality along with the almost total absence of undesirable side effects of dumping, diarrhea, alkaline reflux, weight loss, and nutritional problems. The most consistent criticism of open PGV has been the variable rate of ulcer recurrence from 2% to 30% reported in the literature. The average recurrence rate should be between 2% and 8% in most centers. This variation is often due to technical factors and surgeon variability as well as their ability to complete the PGV effectively. In addition, distal migration of parietal cells into the antrum may also result in recurrence. For most part, the recurrences of the ulcer diathesis can be managed medically; however, in cases where surgery is necessary, truncal vagotomy with antrectomy with either Billroth I or II reconstruction is recommended. Mortality rates below 0.3% are reported. Dumping

and diarrhea have been reported in 3% to 5% of patients but is usually transient. After the operation, the patients may feel some early satiety and upper-abdominal fullness, but these symptoms disappear within a few weeks. Stasis is rare (0.7%) and may be due to either unrecognized pyloric obstruction or complete or near complete antral vagotomy. Jensen demonstrated that the upper 2 to 5 cm of antrum may be denervated without leading to stasis. Stasis that cannot be managed medically often requires reoperation and resection or gastroenterostomy.



CONCLUSIONS

Open PGV is a proven safe and effective operation for the definitive treatment of patients with medically refractory and complicated gastroduodenal ulcer disease in certain clinical settings. Although new paradigms exist today that have revolutionized the treatment of patients with peptic ulcer disease, there still remains a need for occasional definitive ulcer surgery. PGV offers many advantages over truncal vagotomy and antrectomy. It is ironic that the near-ideal surgical procedure finally evolved for the operative treatment of peptic ulcer as the need for such treatment was diminishing.

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9 Laparoscopic Proximal Gastric Vagotomy

Namir Katkhouda and Joerg Zehetner



INDICATIONS/CONTRAINDICATIONS

The objective of the procedure is to divide all vagal nerve fibers innervating acid-producing cells of the stomach while preserving the terminal branches of the main vagal trunks and the nerves of Latarjet, thereby maintaining adequate antral motility. The success of the operation depends on meticulous technique because leaving a single fundic nerve branch intact will allow continued acid secretion in the corresponding gastric secretory zone, leading to early recurrence.

In the elective setting the current indications for the surgical treatment of duodenal ulcer disease are as follows:

- Patients in whom the disease is resistant to medical treatment despite medical therapy for at least 2 years and/or two or more documented recurrences after thorough medical treatment.
- Patients who cannot be followed regularly because of geographic or socioeconomic reasons or who cannot afford medication.
- Patients with complications such as perforation or hemorrhage.

Contraindications for the laparoscopic approach are previous operations on the stomach or upper abdomen with major adhesions and portal hypertension, especially in combination with coagulopathy. Other contraindications are patients with untreated *Helicobacter pylori* infection or patients on nonsteroidal anti-inflammatory drugs before presentation.



PREOPERATIVE PLANNING

The preoperative evaluation of the patient includes evaluation of the general medical status, the operative risk factors, comorbidities, and endoscopic and secretory investigation of the peptic ulcer disease.

The endoscopic examination of the upper gastrointestinal tract is done routinely and allows direct inspection of the diseased area. The duodenal ulcer is typically a linear defect without associated stenosis or hemorrhage. It allows biopsies to rule out possible

H. pylori infection as well as other rare diseases which are associated with duodenal ulcers (Crohn's disease). Other diseases such as esophageal cancer or gastroesophageal reflux disease can be ruled out, as these are diseases with similar symptoms to ulcer disease.

The secretory tests include measurements of basal acid output and peak acid output after stimulation with pentagastrin. These tests are necessary to evaluate the degree of acid hypersecretion in patients who are intractable to medical treatment. The serum gastrin level should always be assessed to exclude gastrinoma.

A radiographic videoesophagram and upper gastrointestinal study can be added if the endoscopy cannot assess a possible postpyloric stenosis.



SURGERY

As with all laparoscopic procedures, this procedure requires general anesthesia and endotracheal intubation. A nasogastric tube is necessary to deflate the stomach before establishing pneumoperitoneum. A urinary catheter is not routinely placed.

Positioning

The patient is positioned in an inverted Y position with the operating surgeon between the legs (French position). The video monitor is positioned above the head and slightly toward the left shoulder of the patient. If the video tower does not provide a flexible screen, the video endoscopic tower is placed on the left of the patient and a second monitor on the right of the patient. The second (camera) assistant stands to the right of the patient and the first assistant and the scrub nurse stand to the left of the patient. The trunk of the patient is elevated 15 degrees in a reverse Trendelenburg position.

Trocar placement

After establishing the pneumoperitoneum with a Veress needle and an optical trocar or with the Hassan technique, first a 12-mm trocar is inserted in the midline approximately one-third of the distance between the umbilicus and the xiphoid process.

A second 12-mm trocar is placed in the left subcostal position in the midclavicular line as a primary working port. Three additional 5-mm or 10-mm trocars are used:

- One in the subxyphoid position used to make a subcutaneous tract for the introduction of the Nathanson Hook Liver Retractor (Automated Medical Products Corporation, Sewaren, NJ). This retractor is secured to the right bed post using an Iron Intern (Automated Medical Products Corporation, Sewaren, NJ).
- Two retracting ports are placed in the right and left subcostal margins.

Technique

In addition to the standard laparoscopic instruments we recommend the following instruments:

- Clip appliers
- Two laparoscopic needle holders
- An L-shaped hook coagulator
- Absorbable monofilament sutures
- Endoloops with preformed Roeder knot
- Application system for fibrin sealant spray
- Ultrasonically activated coagulating shears

There are two major techniques of laparoscopic proximal gastric (highly selective) vagotomy (PGV) for the surgical treatment of chronic duodenal ulcer disease.

Laparoscopic PGV

The procedure begins by the elevation of the left lobe of the liver using a fan retractor or a liver retractor which is inserted via a 5-mm port next to the xyphoid. The surgeon

should be certain that the operation is feasible and that the liver can be retracted so that the operative area is visible. Endoscopic babcock-type clamps provide lateral traction on the greater curvature of the stomach and the lesser omentum is carefully inspected to identify three landmarks: First, the avascular aspect of the lesser omentum, which is crossed by the hepatic branch of the anterior vagus nerve; second, the terminal branch of the anterior vagal nerve (nerve of Latarjet), which runs parallel to the lesser curvature; and third, the terminal “crow’s foot.”

With ultrasonic shears the anterior leaf of the omentum is incised between the nerve and the stomach proximal to the crow’s foot, parallel to the nerve trunk. The lesser omentum is retracted using atraumatic graspers to ensure that the ultrasonic shears divide only the branches of the vagus nerve and not the trunk of the nerve of Latarjet. The identification of individual neurovascular bundles is not necessary. It is important that the blunt jaws of the shears are applied and coapted beyond the vessel to avoid a partial welding that might lead to hemorrhage. Division of the nerves begins at the “crow’s foot” of the nerve of Latarjet. The two most distal branches of the nerve are left intact to be sure that antral and pyloric innervation is preserved. After four to five applications of the shears the dissection then proceeds proximally to include the esophageal branches of the anterior vagus nerve. The fat pad of the cardia containing the anterior nerve trunk is then raised upward and to the right to divide the “criminal” nerves of Grassi near the angle of His. The division of these aberrant branches is of paramount importance to ensure complete vagotomy of the posterior fundus.

The dissection returns to the distal stomach, to retract now the opened anterior leaf of the lesser omentum to expose the posterior vagal branches. The posterior leaf is incised between the stomach and the posterior nerve of Latarjet to open the lesser sac. With the ultrasonic shears all branches of the posterior vagus nerve proximal to the crow’s foot are divided. Using the magnification of the laparoscope, the distal esophagus should be cleared of all nerve fibers for a length of 6 to 8 cm while the two trunks and their terminal gastric branches are preserved.

Alternatively to the technique described with the ultrasonic shears, the anterior and posterior vagal branches can be divided between endoscopic clips with a scissor or with the hook electrocautery (Fig. 9.1).

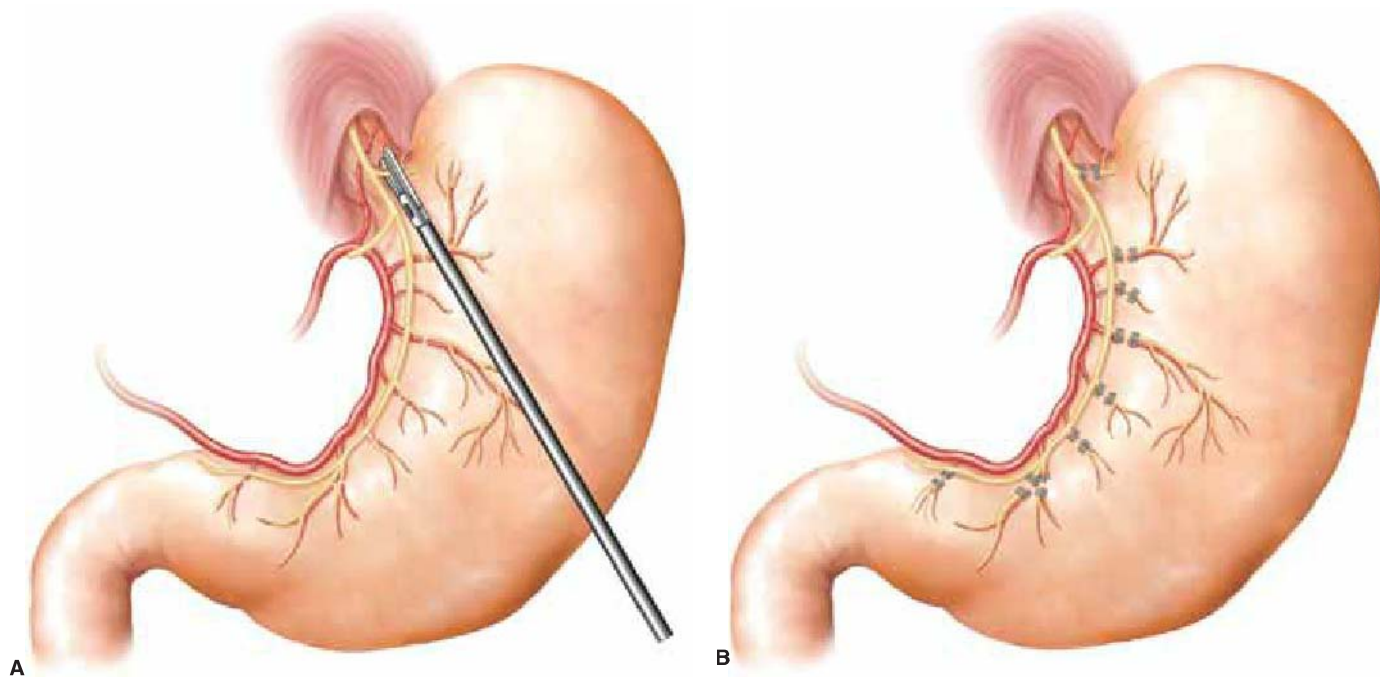


Figure 9.1 The technique of neurovascular ligation in highly selective vagotomy (HSV): (A) laparoscopic technique using harmonic shears, (B) laparoscopic technique using clips.

Laparoscopic Posterior Truncal Vagotomy with Anterior Seromyotomy (Taylor procedure)

This operation is an alternative approach to the previously described technique and was popularized by Taylor and redefined laparoscopically by Katkhouda. The operation has three steps: the hiatal dissection, the posterior truncal vagotomy, and the anterior seromyotomy.

Hiatal Dissection

With a liver retractor or with a probe over the subxiphoid trocar the left lobe of the liver is retracted. The lesser sac is entered through an opening in the pars flaccida. Dissection is continued with ultrasonic sheers, or alternatively the hook cautery, until the muscular portion of the right crus is reached. If left gastric veins are encountered, they may be divided between clips as necessary. Injury to the hepatic branch of the vagus nerve should be avoided if possible, although there is no evidence that division will result in clinically significant postoperative complications.

Posterior Truncal Vagotomy

There are two major landmarks for the posterior truncal vagotomy, which are the caudate lobe and the right crus (Fig. 9.2). With a right-sided forceps the right crus is grasped and held to the right while the ultrasonic sheers or the hook coagulator is used to open the pre-esophageal peritoneum. The abdominal esophagus is retracted to the left, which allows visualization of the areolar tissue. The posterior vagus nerve is identified by its white color. With a grasper the nerve is retracted and adhesions coagulated and divided. The nerve is transected between two clips or with the ultrasonic sheers and a segment of the nerve is removed for histopathology.

Figure 9.2 Posterior truncal vagotomy (important landmarks).

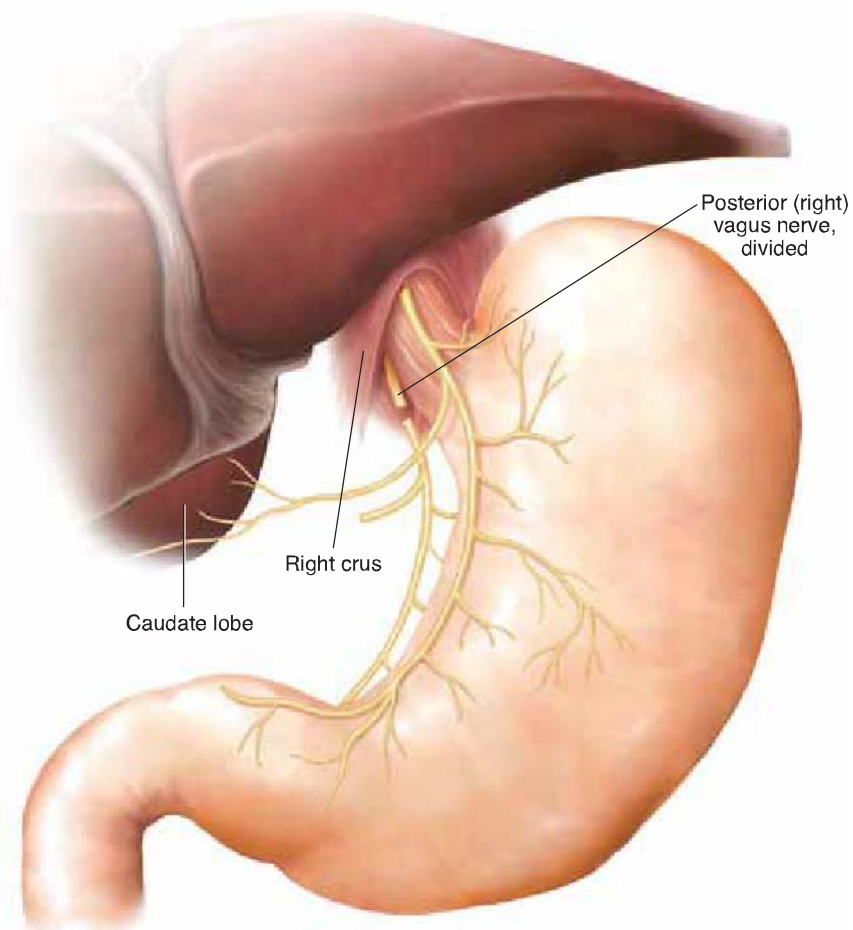
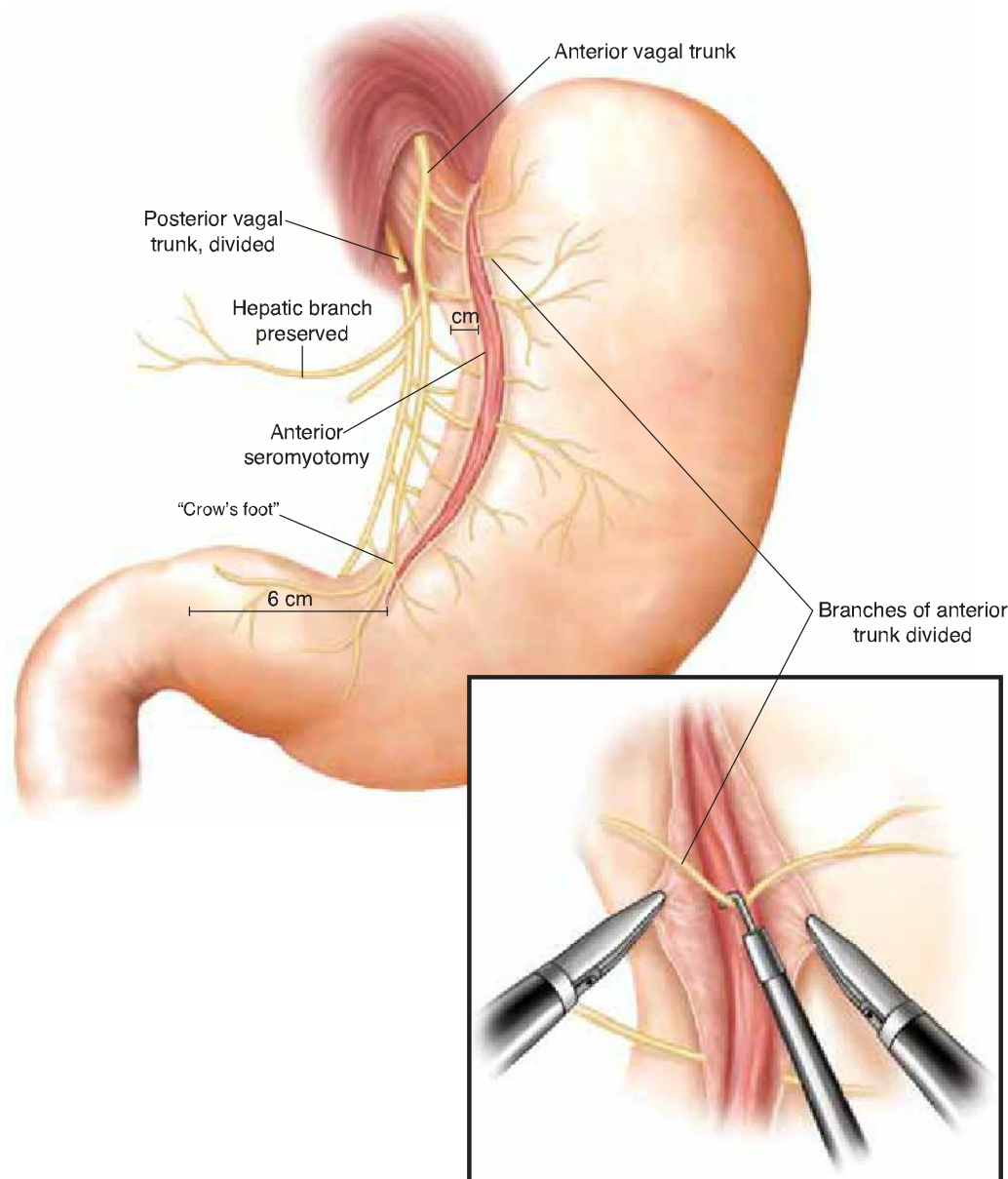


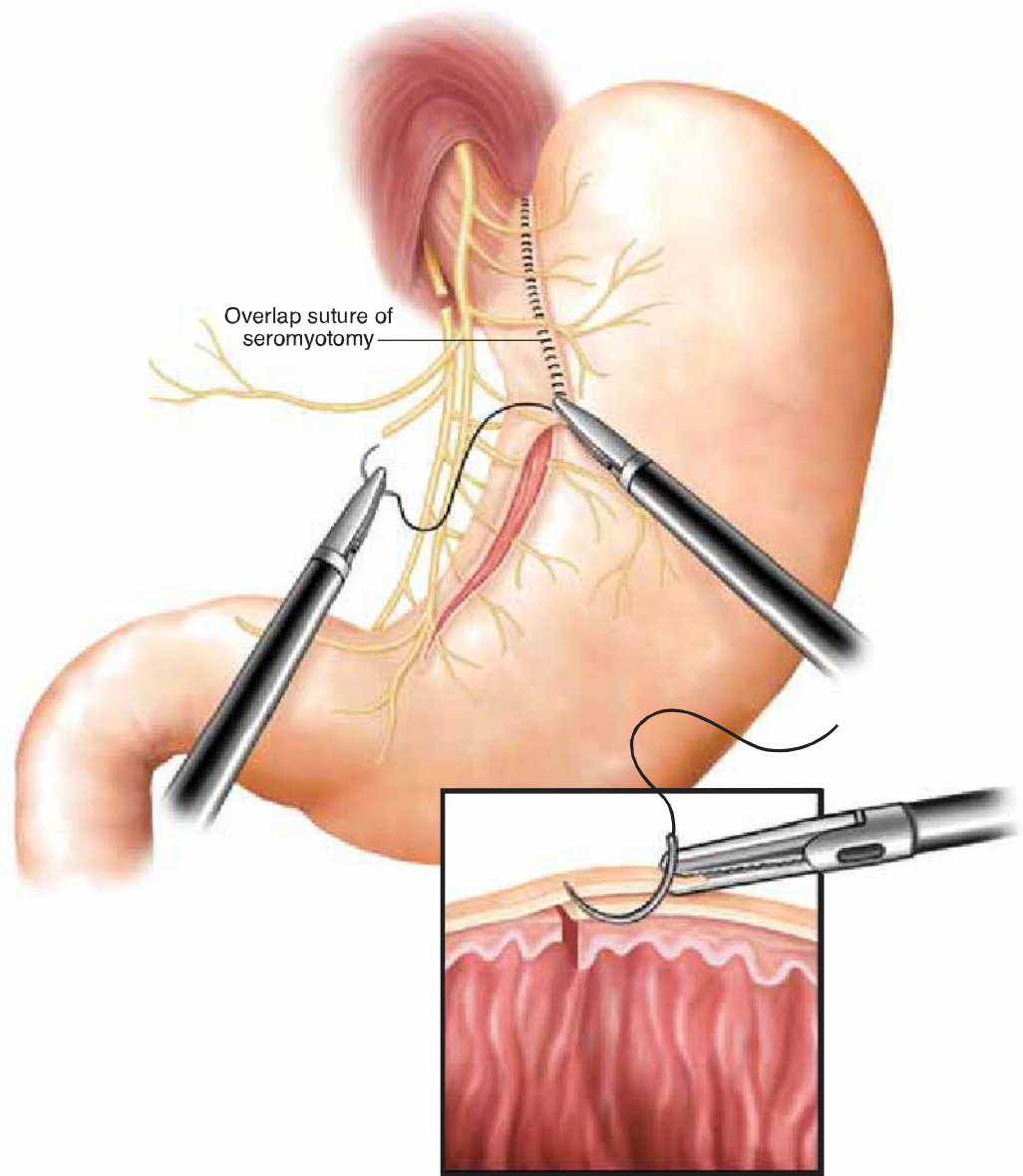
Figure 9.3 Anterior lesser curve seromyotomy.



Anterior Seromyotomy

With two graspers the stomach is spread and the anterior aspect is inspected. With the tip of the electrocautery the line of incision is marked, starting from the gastroesophageal junction parallel to and 1.5 cm from the lesser curvature down to 5 to 7 cm from the pylorus at the level of the crow's foot. The two most distal branches of the nerve are left intact to be sure that the antral and pyloric innervation is preserved. The seromyotomy is then performed with the hook coagulator (monopolar current, blending coagulation, average intensity) along the previously marked line (Fig. 9.3). The hook incises the serosal layer, the oblique layer, and the circular muscular layer. The two borders are grasped and spread apart, which breaks the remaining deep circular fibers. Once the fibers are divided, the mucosa can be easily identified by its typical blue color as it bulges out of the incision. It is very important that the incision is anatomically accurate and hemostasis ensured, and on completion the incision appears 7 to 8 mm in width in the gastric wall. Air is injected through the nasogastric tube to make sure there are no mucosal leaks. Alternatively, the leak test can be performed with methylene blue injected through the nasogastric tube or with an endoscope. The seromyotomy is closed with an overlapping running suture and knotted on both ends (Fig. 9.4). By covering the seromyotomy in a nonanatomic fashion

Figure 9.4 Suture for seromyotomy.



(overlay), postoperative adhesions are prevented; it provides good hemostasis, and nerve regeneration is prevented by the disturbed architecture of the gastric layers.

Other Notes:

The nasogastric tube may be removed at the end of the procedure.

No drainage is placed. Dilation of the pylorus can be performed in the same setting or in a postoperative procedure. A dilation balloon is introduced with a flexible endoscope and moved into the pylorus, inflated to a pressure of 45 psi, and kept in the pylorus for 10 minutes. The success of this procedure is uncertain and might need repetitive sessions to achieve the ideal results.



POSTOPERATIVE MANAGEMENT

The patient is encouraged to ambulate and resume a soft diet on the evening of the operation. Ideally this operation is performed as an outpatient procedure. PPI or H₂ blockers are continued for at least 4 weeks.



COMPLICATIONS

There is a possibility of leaks from the seromyotomy but this has never been experienced by the authors. In the series published by the senior author of this technical review there was no mortality and a low morbidity of 6.6% (mostly pseudo-obstruction with food postoperatively).



RESULTS

The success of the operation can be measured 4 weeks after surgery with pH monitoring. The mean acid reduction is 82%, and in a published series all patients showed healed ulcers on endoscopy. In 3.5% of patients a recurrent ulcer was found on long-term follow-up.



CONCLUSION

The indication for surgery is rare nowadays since the introduction of PPI and H₂ blockers and the successful medical therapy for the eradication of *H. Pylori*. In certain refractory patients this operation is still considered to be the therapy of choice as gastric resection should not be performed for benign cases such as duodenal ulcers, unless a stricture makes it necessary to perform a subtotal gastrectomy.

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10 Patch for Perforated Ulcer

David W. Mercer and Matthew R. Goede



INDICATIONS/CONTRAINDICATIONS

Patients with perforated peptic ulcers usually present with acute onset of abdominal pain. Patients frequently report that the pain started “as if someone flipped a switch.” However, elderly patients frequently present with a less robust onset of pain. Occasionally, patients will complain of left shoulder pain from diaphragmatic irritation and some will complain of nausea and vomiting. While some patients report a history of recent gastritis, frequent steroid or nonsteroidal anti-inflammatory drug use, or use of cocaine, perforation may be the first clinical sign that patients have underlying peptic ulcer disease. On radiographic imaging, pneumoperitoneum is present in greater than 80% of patients.

Nonoperative management of perforated ulcers has been described since the 1940s; however, its use and patient selection remains a major debate. Patients who present with pneumoperitoneum who have a water-soluble contrast gastroduodenogram showing no extravasation, no signs of septic shock, and abdominal examination findings confined to the upper abdomen may be considered for nonoperative treatment. Nonoperative treatment routinely includes nasogastric decompression, antibiotics to cover enteric pathogens, antibiotics to cover *Helicobacter pylori* if present, and proton pump inhibitors. However, approximately 30% of patients treated nonoperatively do not improve and require surgical intervention. Patients treated nonoperatively also have about a 30% longer inpatient length of stay. Nonoperative management also has an extremely high failure rate (around 70%) in the elderly, patients with comorbidities, patients with hemodynamic instability, perforations greater than 48 hours old, and in patients with impaired wound healing. This creates a dilemma because the patients that you would prefer to treat nonoperatively are the ones most likely to fail nonoperative therapy. Therefore, operative treatment is the preferred modality except in an extremely selective subset of patients.

The classic indication for use of a patch for perforated ulcer is in the unstable patient with shock, the presence of extensive peritonitis or abdominal abscess formation, or patients with severe comorbidities. While patch repair of perforated ulcer was described as a damage control technique, with the development of effective medical therapy for ulcer disease and the corresponding decrease in patients requiring surgical therapy for ulcer disease, more surgeons are utilizing patch repair in the acute setting. In addition, the average age of patients presenting with perforated duodenal ulcers is

increasing, with the average age now between 65 and 75 years. With this older age at presentation, it is more likely that these patients will have comorbidities that lead the surgeon to opt for a less involved procedure in the acute phase.



PREOPERATIVE PLANNING

While a large majority of patients do not present in hemodynamic shock, appropriate preoperative resuscitation is still needed. Placement of a nasogastric tube to decompress the stomach and limit peritoneal contamination should be performed. Antibiotics to cover enteric pathogens should be administered, along with appropriate goal-directed crystalloid resuscitation. If possible, a determination of the presence of *H. pylori* through a serum antibody test should be done to aid in operative planning. Urea breath tests should not be done in patients with suspected perforation. In addition, the operative plan should be decided based on the patient's presentation. Patients who present with hemodynamic instability, signs of end-organ failure, or shock should be classified as unstable.

- Gastric ulcer
 - Unstable patients should be treated with patch repair and biopsy of ulcer or ulcer resection depending on the location of gastric ulcer.
 - Patients with known *H. pylori* may be treated with patch repair and *H. pylori* eradication and biopsy.
 - In medication-induced ulcers, in which the medication regimen can be altered, patients may be treated with patch repair and biopsy.
 - Stable patients who are known to be *H. pylori*-negative, patients with a long history of peptic ulcer disease, or patients with essential medication needs (chronic steroid use) should undergo a definitive antiulcer procedure.
- Duodenal ulcer
 - Unstable patients should be treated with patch repair.
 - Initial presentation of duodenal ulcer disease, especially if *H. pylori* is present, can be treated with patch repair, proton pump inhibitors, and *H. pylori* eradication.
 - Stable patients with ulcer disease refractory to proton pump inhibitors and *H. pylori* eradication should be treated with a definitive antiulcer procedure.

In patients who are profoundly septic, are hypotensive, or on vasopressors, one must carefully consider if a patient can tolerate pneumoperitoneum before proceeding with a laparoscopic repair. Also in patients with extensive previous abdominal surgery, significant duodenal scarring, or concomitant bleeding ulcer, the use of laparoscopy may be limited. Given the intra-abdominal inflammatory process caused by the perforation, the technique for obtaining initial intra-abdominal access may need to be modified.



SURGERY

Open Technique

- Patient is placed supine.
- Sequential compression devices and urinary catheter are placed.
- Abdomen is entered through an upper midline incision to allow for exploration of entire abdomen if gastric or duodenal perforation is not the cause of peritonitis.
- Following repair of the perforation, a thorough irrigation of the abdomen with attention paid at the right and left subphrenic spaces and pelvis should be done.

The laparoscopic approach is becoming more widely used in these patients. Laparoscopy offers several advantages, most notably the magnification provided by the laparoscope, less postoperative pain, fewer postoperative pneumonias, and fewer wound

complications. Also laparoscopy affords the ability to examine the entire abdomen without extending or relocating incisions, which is especially helpful when the exact location of the perforation has not been identified preoperatively. It also aids in performing a thorough inspection and irrigation of the entire abdomen under direct visualization.

Patch Techniques

Frequently any omental reinforcement of an ulcer perforation is referred to as a Graham patch. The original nonpedicled patch (Fig. 10.1A) described by Graham is infrequently used today in favor of pedicled omental flaps. Maintaining a vascular pedicle allows the omentum to remain alive and continue in its natural function as “watchdog” of the abdomen.

- Primary closure with omental pedicle flap (Fig. 10.1B)
 - Used in small ulcers where the tissue has minimal inflammation and can be brought together easily, without narrowing the duodenum.
 - Edges of ulcer are debrided if necessary, and the defect is closed primarily in a transverse orientation, usually in a single layer using 2-0 or 3-0 silk.
 - A vascularized pedicle of omentum is brought up to cover the repair.
 - The pedicle is secured in place with the long tails of the ulcer closure sutures or separate tacking sutures of 3-0 silk around the margins of the primary closure.
- Omental pedicle flap (Cellan-Jones repair) (Fig. 10.1C and Fig. 10.2)
 - Excise friable edges.
 - Send the gastric specimen for pathologic examination.
 - Classically place four to six 3-0 sutures through the viscera wall, transversing the ulcer.
 - Anchor an omental pedicle with a single suture beyond the ulcer.
 - Tie the previously placed sutures down like an “archway” over the length of the pedicle thereby securing the omentum firmly within the edges of the perforation.
 - The lack of primary closure decreases the risk of narrowing the duodenum in the case of a large perforation.

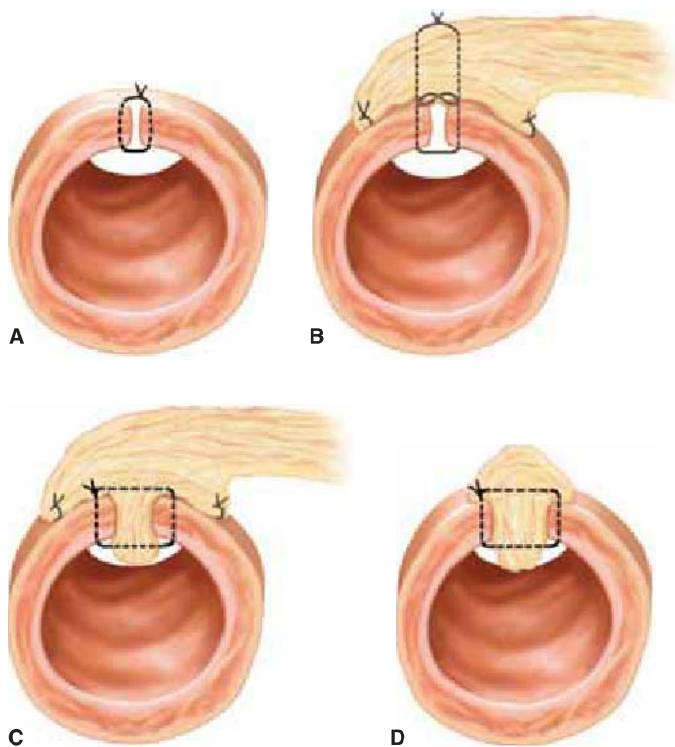


Figure 10.1 A–D: Different techniques to apply omental patch to close perforated ulcer.

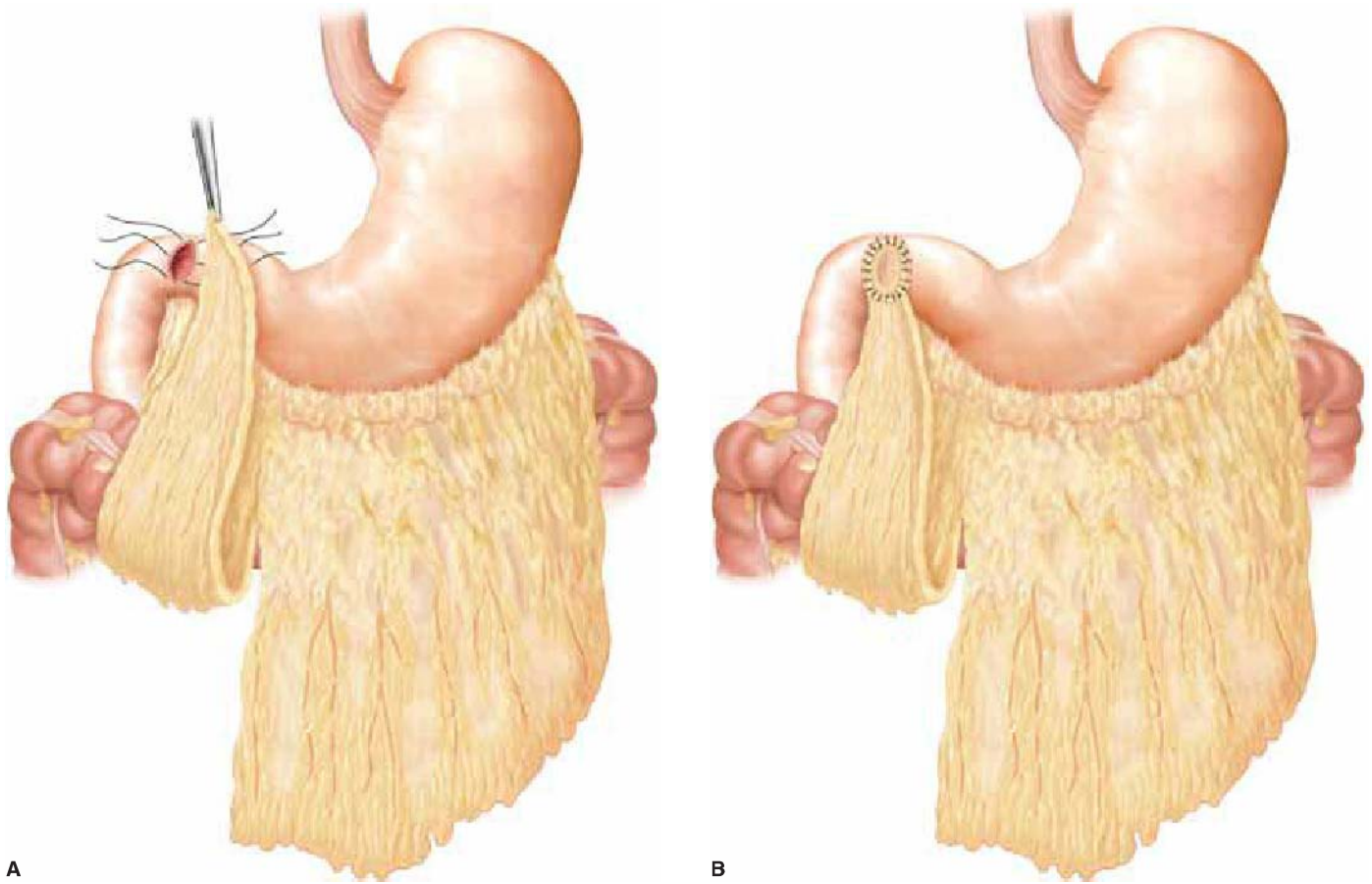


Figure 10.2 A and B: Different techniques to secure omental patch using long tails or tacking sutures.

- Free omental plug (Graham patch) (Fig. 10.1D)
 - The lack of primary closure decreases the risk of narrowing the duodenum in the case of a large perforation.
 - Place approximately three sutures parallel to the long axis of the duodenum across the ulcer.
 - Place a piece of free omentum within the ulcer defect and tie the sutures over the top of the free graft to secure the omentum to the ulcer defect.
 - Relies on the free piece of omentum to serve as a scaffold on which the body can secrete fibrin.
- Serosal patch
 - Used in large perforations, frequently following traumatic duodenal injuries
 - A loop of jejunum is brought up to cover the perforation and is sewn circumferentially to the edges of the ulcer once they have been debrided in a fashion similar to a small bowel anastomosis; however, the bowel is not opened.

Operative Pearls

- Unless excised, gastric ulcers should always undergo biopsy prior to closure or patch repair. These biopsies are essential given the risk of gastric cancer leading to perforation or the presence of *H. pylori* infection involved with perforation.
- The sutures used to secure the omental patch should not be overly tightened, as doing such would compromise the vascular supply to the patch.
- The repair can be tested either by filling the upper abdomen with saline and then instilling air through the nasogastric tube or by the placement of methylene blue dye down through the nasogastric tube.

- Placement of a nasojejunal tube should be considered prior to repair of large ulcers to allow the possibility of enteral feeding beyond perforation.
- In elderly or debilitated patients or in patients with a high concern of possible continued leak, placement of a jejunostomy tube for distal enteral feedings should be considered if the patient is tolerating anesthesia.



POSTOPERATIVE MANAGEMENT

Inpatient

- While the duration of nasogastric tube decompression remains controversial, many surgeons promote at least a brief period of gastric decompression to allow time for the patch to fully adhere to the visceral wall.
- Drains are rarely left anymore; however, if placed, drains should be closely monitored for the presence of bile. The peritonitis associated with a perforated peptic ulcer frequently leads to a significant amount of fibrin deposition, which makes drains nonfunctional after just a day or two.
- Tachycardia may be the earliest sign of developing sepsis or a continued leak from the repair.
- Patients should be started on intravenous proton pump inhibitors postoperatively and converted over to oral once tolerating a diet.

Outpatient

- *H. pylori* infection is present in 80% of gastric ulcer perforations and 90% of duodenal perforations. All patients should undergo evaluation for and treatment of *H. pylori* infection if present following recovery from their perforation and discharge from the hospital.
 - If follow-up endoscopy is planned to evaluate for ulcer healing (required in gastric ulcers to rule out malignancy), endoscopic biopsies of the gastric antrum can be performed to evaluate for the presence/eradication of *H. pylori*.
 - Urea breath tests and stool antigen tests can also be done to detect *H. pylori* infection and determine whether it has been eradicated.
- If *H. pylori* is detected, appropriate triple therapy including proton pump inhibitor and antibiotics should be administered and eradication again confirmed.
 - If eradication is not achieved, repeat treatment with quadruple therapy should be considered.



COMPLICATIONS

Continued leak has been reported in 5% to 10% of patients treated with omental patch repair. Patients with suspicion of a continued leak should be evaluated by a water-soluble contrast study. If the leak is small or contained, one could entertain treating the collection with percutaneous techniques; however, the possibility exists of developing a persistent fistula, which, depending on the patient's condition, could be a desirable outcome over reoperation.

If the persistent leak is from a gastric ulcer, a wedge resection may be possible. However, if one is going to reoperate on a patient, a definitive antiulcer procedure should be attempted unless the patient remains septic or the degree of inflammation or contamination is prohibitive.

If the persistent leak is from a duodenal ulcer, again a definitive ulcer operation needs to be strongly considered. In a septic patient, triple tube therapy including a gastrostomy for proximal drainage, a jejunostomy with the tube fed retrograde to drain

the duodenum near the perforation, and a distal jejunostomy for enteral feeding should be placed. A duodenostomy should be avoided to drain the duodenum as this will limit future attempts at definitive surgical treatment.

Intra-abdominal abscess formation is possible and can usually be treated with radiographically guided percutaneous techniques. In the past, abdominal irrigation through drains left at the initial operation was attempted, which did not decrease abscess formation and significantly increased demands on the healthcare providers to monitor the continuous input and output. Likewise, the routine placement of drains in the peritoneal cavity has not decreased the incidence of abscess formation.

Sepsis is frequently the complication that leads to death in these patients. Most of these patients warrant at least brief observation in an intensive care unit postoperatively, and healthcare providers must vigilantly observe for signs of sepsis even when the patient is sent to the general surgical ward. Empiric antibiotics with a broad spectrum should be continued depending on the degree of peritoneal contamination. Empiric antifungal coverage should be considered as well given the proximal nature of this visceral perforation. Patients who develop overwhelming sepsis should be started on antifungal coverage, as frequently by the time the fungus is isolated from blood or peritoneal cultures the patient has already succumbed to the sepsis.



RESULTS

- The mortality of perforated peptic ulcers is around 3% for healthy, younger patients.
- Unfortunately the mortality rate can approach 30% in elderly and debilitated patients.
- In perforated duodenal ulcers the use of patch repair followed by effective medical acid suppression and *H. pylori* eradication has a 1-year recurrence rate of around 5%.



CONCLUSIONS

The use of patch repair for perforated ulcer is more frequent now than in past years. The introduction of effective acid-suppressing medications, a better understanding of the role of *H. pylori* and its eradication, introduction of minimally invasive techniques, and the decreasing experience with acid-reducing procedures have led many surgeons to depend on patch repair as their technique of choice in the acute setting.

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11 Laparoscopic Patch of Perforated Duodenal Ulcer

Sunil Sharma and Bestoun H. Ahmed



INDICATIONS/CONTRAINDICATIONS

The success of medical management of peptic ulcer disease has reduced the overall need for surgical intervention. But, the incidence of perforation, which is a serious and potentially fatal complication, remains the same (5% to 10%). The surgical intervention of simple closure of the perforation with or without omentoplasty is accepted in most of these patients. Modern antiulcer therapy (i.e., H_2 receptor antagonists or proton pump inhibitors [PPI] with anti-*Helicobacter pylori* agents) has decreased the high recurrence rate from 40% to 3–4%.

Laparoscopic patch of perforated duodenal ulcer has become an alternative method to the open procedure since its initial report in 1989 by Mouret. It offers advantages of better visualization of the peritoneal cavity, thorough lavage and lack of upper abdominal incision with its related complications, especially in high-risk patients. The need for definitive surgical procedures is diminishing. However, it depends on the chronicity and intractability of the disease.

Laparoscopic closure is a favorable method in most patients. It is especially indicated in

- those with age less than 70 years,
- symptoms persisting less than 24 hours,
- absence of shock on admission,
- American Society of Anesthesiologists grade less than III, and
- perforation smaller than 10 mm in diameter (Fig. 11.1).

There are no absolute contraindications for the laparoscopic approach. It is preferable to avoid this approach in patients with Boey's score of 2 or 3. Patients with such a score range are associated with high morbidity and mortality. Boey score (0 to 3) is the count of Boey risk factors: shock on admission, American Society of Anesthesiologists grade III to V, and prolonged perforation. A Boey score of 0 means low surgical risk; otherwise, it is considered a high surgical risk.

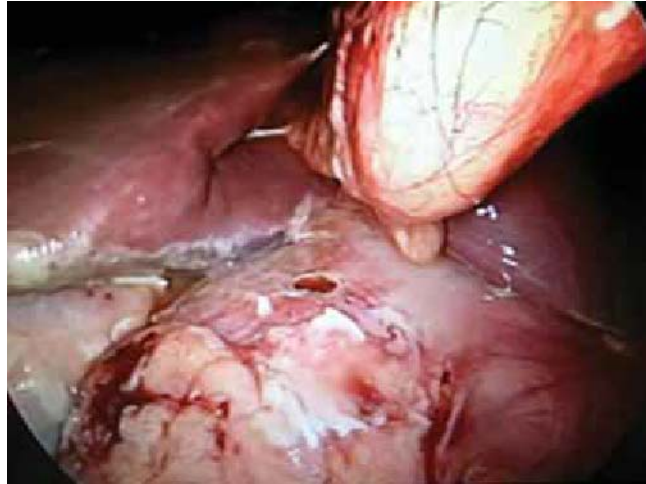


Figure 11.1 Perforated duodenal ulcer. (Image courtesy of D. Oley-nikov, University of Nebraska Medical Center.)



PREOPERATIVE PLANNING

- Diagnosis is achieved by the history of sudden onset of abdominal pain with abdominal findings indicating peritonitis. Free air under the diaphragm is expected in 85% of patients. Usually, no further diagnostic measures are indicated.
- Following diagnosis, the preoperative resuscitation is achieved by intravenous fluids (crystalloids), nasogastric intubation, and urinary catheterization.
- Therapeutic intravenous antibiotics should be started preoperatively and continued postoperatively. Once the patient tolerates oral intake, antibiotics will be switched to an oral formula which treats *H. pylori* organisms.
- H₂ blockers or PPI should also be started preoperatively and continued throughout the postoperative period.
- These patients must be scheduled for urgent laparoscopic exploration.
- Diagnostic laparoscopy is indicated in all patients except the minority who present with a high Boey's score. For this small group of patients, starting with exploratory laparotomy is the safest approach.



SURGERY

Positioning

- The patient should be in supine position with sequential compression devices and safety belts on the lower extremities. Both arms should be tucked to allow freedom of movement for the surgeon and assistants.
- Some surgeons prefer to stand between the patient's legs while others perform the procedure from the patient's left side.
- Standing on the patient's right side makes suturing an easier task. But, change of position may be needed to achieve an efficient peritoneal lavage (Fig. 11.2).



SURGICAL TECHNIQUE

- Under general anesthesia, the patient is placed in a 15- to 20-degree reverse Trendelenburg position. The operating surgeon stands on the patient's left. This initial position allows better diagnostic exploration and more efficient peritoneal lavage. An

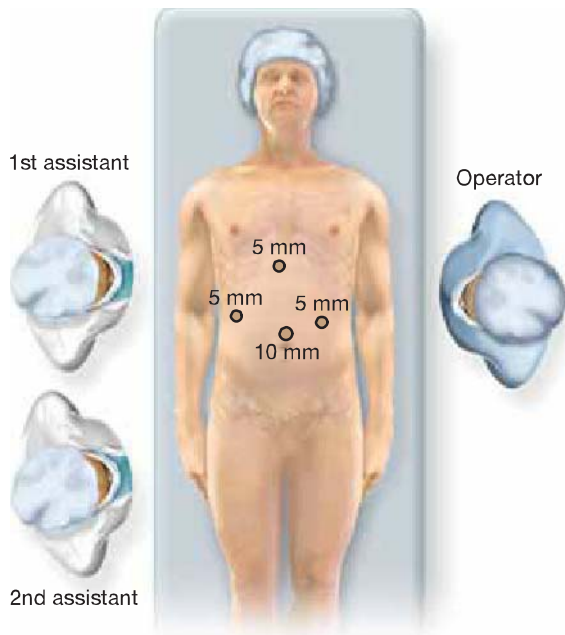


Figure 11.2 An example of port site placement.

assistant stands on the right side. Later the surgeon may switch the position to the right side for easier suturing.

- All port sites should be infiltrated both subdermally and preperitoneally with 0.5% bupivacaine prior to incision or creation of the trocar path.
- Through a 2-mm-long left hypochondrial stab incision just inferior to the costal margin at the midclavicular line, the carbon dioxide pneumoperitoneum (up to 15 mm Hg) is established with a Veress needle.
- After establishing the pneumoperitoneum, through a 10-mm-long supraumbilical curvilinear incision a 10-mm trocar is placed and an angled laparoscope with 30° or 45° is introduced and the entire abdominal cavity is thoroughly investigated. If the laparoscopic findings correspond with the preliminary diagnosis of perforated ulcer, additional trocars are placed under laparoscopic view.
- A 5-mm port is placed one hand-breadth below the left costal margin at the midclavicular line. This port is used as the right-handed working port and can be used to introduce the ski-needle and suture or could be exchanged for a 12-mm port if a regular curved needle or an Endostitch (Covidien, Norwalk, CT, USA) is used. Another 5-mm port is placed in the right hypochondrium at the midclavicular line one hand-breadth from the liver edge on the abdominal wall below the costal margin. This port is the left-handed working port.
- If the liver is covering the perforation site, a 5-mm trocar is placed in the midepigastic region and used for liver and gallbladder retraction (Fig. 11.3).



Figure 11.3 The liver is covered by bile-stained fibrinous membrane.



Figure 11.4 Laparoscopic primary closure of perforated duodenal ulcer. (Image courtesy of D. Oleynikov, University of Nebraska Medical Center.)

- After the ulcer size is carefully measured with reference to the 5-mm-diameter working laparoscopic instrument, the perforation is closed with double or triple Vicryl or silk sutures (Fig. 11.4). The perforation is closed by intracorporeal knotting with an omental patch (omentopexy) directly applied to the perforation site (Modified Graham's technique). This approach avoids tension on the inflamed ulcer margin and cutting through the tissue (Figs. 11.5 and 11.6).
- It is critical to avoid anchoring the posterior duodenal wall when these sutures are applied. The best approach is to pull the needle through the perforation and reinserting it through the perforation to complete the next half of the stitch.
- Giant duodenal ulcer perforations (greater than 2 cm in diameter) are better to be dealt with by open exploration and definitive acid reduction operation. Great care should be practiced to avoid duodenal narrowing when closing larger perforations.
- Intracorporeal knotting is the preferred method. Extracorporeal suturing technique could be used for small ulcers with little inflammation. This should be done with care to avoid cutting through the friable ulcer margin.
- The air–fluid leak test is optional as this may lead to overdistension of the bowel.
- Thorough peritoneal lavage is another key intervention of the operation. This is performed using 3 to 5 L of warm normal saline with the patient in various positions. Special attention is given to potential spaces for collection. These are bilateral sub-diaphragmatic, subhepatic, lateral paracolic gutters, and pelvic cavities.
- Drainage of the peritoneal cavity is not necessary.
- All ports larger than 5 mm are closed with the help of a transabdominal suture passer device using 0-Vicryl suture material to close the fascia.

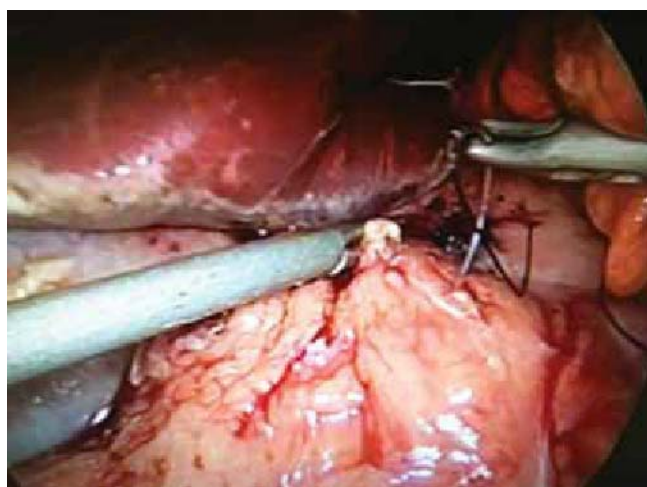


Figure 11.5 Placing tacking sutures for patch repair of perforated duodenal ulcer. (Image courtesy of D. Oleynikov, University of Nebraska Medical Center.)

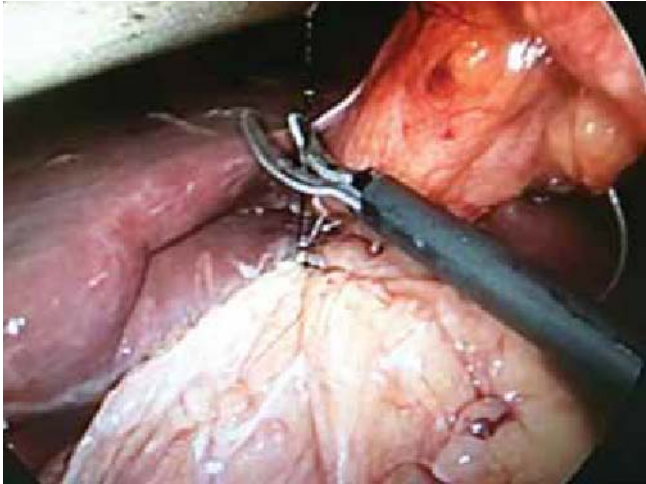


Figure 11.6 Completed pedicled omental patch of perforated duodenal ulcer. (Image courtesy of D. Oleynikov, University of Nebraska Medical Center.)

Alternative Techniques

Alternative methods have been described to close the perforation if suture repair is challenging. Examples of these methods are as follows:

- Closure of the perforation by a gelatin sponge glued into the perforation
- The perforation is closed by fibrin glue.
- The automatic stapler has been used for perforation site closure.
- The simple “one-stitch” repair with omental patch.
- Use of running suture to avoid intracorporeal or extracorporeal knotting.
- Combined laparoscopic–endoscopic repair.

Definitive Ulcer Surgery

- The policy of simple closure alone without definitive procedures is sufficient, provided it is followed by anti-*H. pylori* antibiotics and PPI management.
- This is especially true for patients who are poor surgical risks with severe peritonitis.
- Performing an elective definitive procedure is now restricted to
 - those patients with a long history of dyspepsia (more than 3 months) with rapid recurrence of symptoms following discontinuation of H₂ blockers or PPI;
 - those with true recurrence
 - those with other complications of ulcer (e.g., stricture);
 - Giant duodenal ulcer.
- Definitive procedures such as posterior truncal vagotomy and anterior highly selective vagotomy or anterior seromyotomy can be performed laparoscopically.



POSTOPERATIVE MANAGEMENT

- All patients receive intravenous fluids, nasogastric tube decompression, and parenteral analgesics.
- The nasogastric tube should be removed after 24 to 48 hours according to its output.
- Upper GI study to confirm absence of leak is optional and depends on the magnitude of inflammation and delay in presentation.
- Oral intake is started in general after 48 hours, beginning with clear liquids and advanced accordingly. This depends on the degree of peritonitis and hence the duration of bowel recovery.

- Perioperative administration of intravenous antibiotics to cover bowel organisms is essential as well as intravenous PPI or H₂ blockers.
- Once oral intake is established, the triple therapy regimen is initiated which consists of clarithromycin and amoxicillin for 14 days, in addition to a PPI for 4 weeks.
- This is followed by esophagogastroduodenoscopy after 2 months to check the healing status of the ulcer and existence of persistent *H. pylori* organisms.



RESULTS

- The incidence of perforated duodenal ulcer has not been reduced comparably despite the overall decline in the incidence of complicated peptic ulcer disease. That may be due to the widespread use of nonsteroidal anti-inflammatory drugs in the last three decades.
- Laparoscopic closure of the perforation offers important advantages:
 - Decreasing postoperative pain
 - Less abdominal wall complications
 - Improved quality of visualization and peritoneal lavage.
- Some argue against the laparoscopic choice as being a time-consuming procedure, in addition to the equipment costs; however, the operative time decreases with experience.
- No significant difference has been found in the time of postoperative recovery and hospital stay in comparison to the open method (laparotomy), which is probably due to the degree of peritonitis and intestinal ileus. But, it leads to earlier return to the previous lifestyle.



COMPLICATIONS

- The pooled estimate of the effect in the meta-analyses of multiple randomized controlled trials showed that the incidence of complications was no higher after laparoscopic (17.2%) than open repair (17.8%), $p = 0.50$.
- Comparing the laparoscopic group to the open approach, the incidence of complications are as follows:
 - Suture-site leakage after using laparoscopic omental patch repair is statistically comparable to the traditional laparotomy repair.
 - The incidences of thoracic complications (pneumonia and pleural complications), prolonged postoperative ileus, and intra-abdominal abscess formation were no different.
 - The wound complications rate: the pooled estimate favored laparoscopic over open repair.
 - Meta-analysis showed a higher reoperation rate after laparoscopic repair, but the difference was not significant ($p = 0.10$).
 - The pooled estimate of the mortality rate favored laparoscopic repair and the difference was statistically significant ($p < 0.001$). However, meta-analysis of randomized controlled trials did not demonstrate any significant reduction in mortality following laparoscopic repair ($p = 0.63$).



CONCLUSIONS

- In case of suspected perforated duodenal ulcer, laparoscopy should be advocated as a diagnostic and a potential therapeutic tool.
- Laparoscopic omental patch repair with intracorporeal knotting is the preferred approach.
- Triple anti-*H. pylori* and PPI therapy is required in the postoperative period.
- Closure of perforated duodenal ulcer allows early return to normal lifestyle and has a lower incidence of abdominal wall complications.

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12 Ligation Bleeding Ulcer, Vagotomy, Pyloroplasty

Eric S. Hungness



INDICATIONS

Despite the significant decrease in the surgical treatment of complicated peptic ulcer over the past two decades, duodenal ulcer remains one of the most common sources of upper gastrointestinal (UGI) bleeding. The incidence of bleeding from duodenal ulcers that is severe enough to necessitate emergency endoscopic or operative intervention also has not decreased over the past decade.

Patients may present with hematemesis or melena in varying degree of hypovolemic shock. If the patient is unstable, he or she should be taken to an intensive care unit and resuscitated immediately. Resuscitation of an unstable patient is begun by establishing a secure airway and ensuring adequate ventilation. Oxygen should be given, with a low threshold for endotracheal intubation. Isotonic crystalloid is infused after reliable intravenous (two large bore, short peripheral catheters) catheters are placed. For patients in class 3 or 4 shock or those with poor peripheral access, a single lumen 8 French central line should be considered. A urinary catheter should be inserted with close monitoring of urine output. A complete blood count, blood chemistries (including tests of liver function and renal function), and measurement of the prothrombin time and the partial thromboplastin time should be drawn, and a specimen should be sent to the blood bank for typing and crossmatching.

Attempts to establish the cause of bleeding should only be attempted after the initial measures to protect the airway and stabilize the patient have been completed. The next step is nasogastric (NG) aspiration. A bloody aspirate is an indication for esophagogastroduodenoscopy, as is a clear, nonbilious aspirate if a bleeding site distal to the pylorus has not been excluded. The source of the bleeding is unlikely to be the stomach or duodenum with a nonbloody, bile-stained aspirate. Nonetheless, if subsequent evaluation of the lower GI tract for the source of the bleeding is unrewarding, a duodenal ulcer that had stopped bleeding prior to NG tube passage should still be considered. Esophagogastroduodenoscopy should then be performed to confirm the presence of a duodenal ulcer. Most bleeding duodenal ulcers may be controlled endoscopically, although the degree of success to be expected in individual cases varies according to the expertise of the endoscopist.

In general, a 6-unit blood transfusion or bleeding that is not able to be controlled endoscopically mandates surgical intervention. Likewise, ongoing hemorrhage in a hemodynamically unstable patient (especially an elderly one) calls for immediate surgical therapy. The following endoscopic findings for patients whose bleeding is controlled endoscopically should also be strongly considered for surgical therapy: (1) a visible vessel, (2) active bleeding, (3) adherent clot, and (4) giant ulcers.

If the bleeding is controlled endoscopically, intravenous proton pump inhibitor therapy should be given either in a bolus twice daily or by continuous infusion. Antibiotics directed against *Helicobacter pylori* (e.g., a 14-day course of metronidazole, 500 mg p.o., t.i.d.; omeprazole, 20 mg p.o., b.i.d.; and clarithromycin, 500 mg p.o., b.i.d.) should be administered if the organism is present, as treatment has been shown to reduce rebleeding rates. If bleeding recurs, a second attempt at endoscopic control should be made. Repeat endoscopic treatment reduces the need for surgery without increasing the risk of death and is associated with fewer complications than surgery.

For patients who do require urgent operative intervention from a bleeding duodenal ulcer, duodenotomy, suture control of hemorrhage, and subsequent pyloroplasty and truncal vagotomy allows for best balance between expeditious and effective treatment. Some have advocated selective vagotomy or highly selective vagotomy to reduce postvagotomy syndromes; however, selective vagotomy affords little advantage over truncal vagotomy. Moreover, highly selective vagotomy is a procedure that most surgeons are unfamiliar with and requires significant expertise to effectively reduce the chance of rebleeding. The proximal (truncal) vagus nerve anatomy is more consistent resulting in the fact that truncal vagotomy is more easily reproduced and expeditious when performed by the majority of surgeons.



PREOPERATIVE PLANNING

There is limited preoperative planning as these cases are usually performed in an urgent or emergent basis. All patients who present with UGI bleeding should routinely have

- type and cross for 6 units,
- two sites of reliable intravenous access,
- correction of coagulopathy.



SURGERY

Positioning

Patients should be placed in a supine position and prepped and draped in standard fashion for general anesthesia so that the anesthesia team has easy access to the IV lines. A reliable arterial line should be placed for continuous blood pressure monitoring and arterial blood gas analysis. These patients are prone to relative hypothermia due to the large volume of crystalloid and blood products infused, thus an upper or lower Bair hugger should be placed and fluid warmers used as necessary. Mechanical compression devices are applied for thromboprophylaxis, and patients should receive appropriate preoperative IV antibiotics. A Bookwalter or Omni retractor is usually necessary for these cases, so having one arm tucked may be useful. A functional NG tube and urinary catheter should have been placed preoperatively.

Exposure

An upper midline celiotomy is made from the xiphoid to just above the umbilicus. This usually provides adequate exposure for both parts of the operation. It is often necessary to extend the incision proximally to the left of the xiphoid to gain adequate exposure

to the hiatus. A Bookwalter or Atlas retractor is usually required, particularly for the vagotomy portion of the operation to elevate the rib cage. It may also be necessary to take down the triangular ligament and mobilize the left lobe of the liver to optimize hiatal exposure. Care must be taken with adequate padding to retract the left lobe medially and avoid liver laceration.

Duodenotomy and Ligation of Bleeding Vessel

After the abdomen is carefully inspected and any adhesiolysis completed, retractors are placed to optimize exposure of the distal stomach and duodenum. Appropriate positioning of the NG tube is confirmed. After the pylorus is located, two 3-0 traction sutures are placed cephalad and caudad approximately 1 cm apart on the pylorus muscle, and a 6-cm enterotomy is made to expose the posterior bleeding ulcer (Fig. 12.1). Typically, the well-described “vein of Mayo” can help identify the pylorus. Manual finger pressure can temporarily control active bleeding to allow the anesthesia team to restore the intravascular volume in patients with hypovolemic shock. The bleeding vessel is usually from the gastroduodenal artery that runs cephalad to caudad or a branch of the gastroduodenal artery that may have a more medial course toward the pancreas. Therefore, a superior, inferior, and “U” stitch should be placed to ensure adequate vascular control (Fig. 12.2). Care must be made not to take excessively deep bites as to avoid injury to the common bile duct. Once the bleeding is stopped, attention is placed on closing the duodenotomy with a Heineke–Mikulicz pyloroplasty.

Heineke–Mikulicz Pyloroplasty

The pyloroplasty is started by placing the initial traction sutures on tension to create the cephalad and caudad corners of the transverse closure (Fig. 12.3). The full thickness

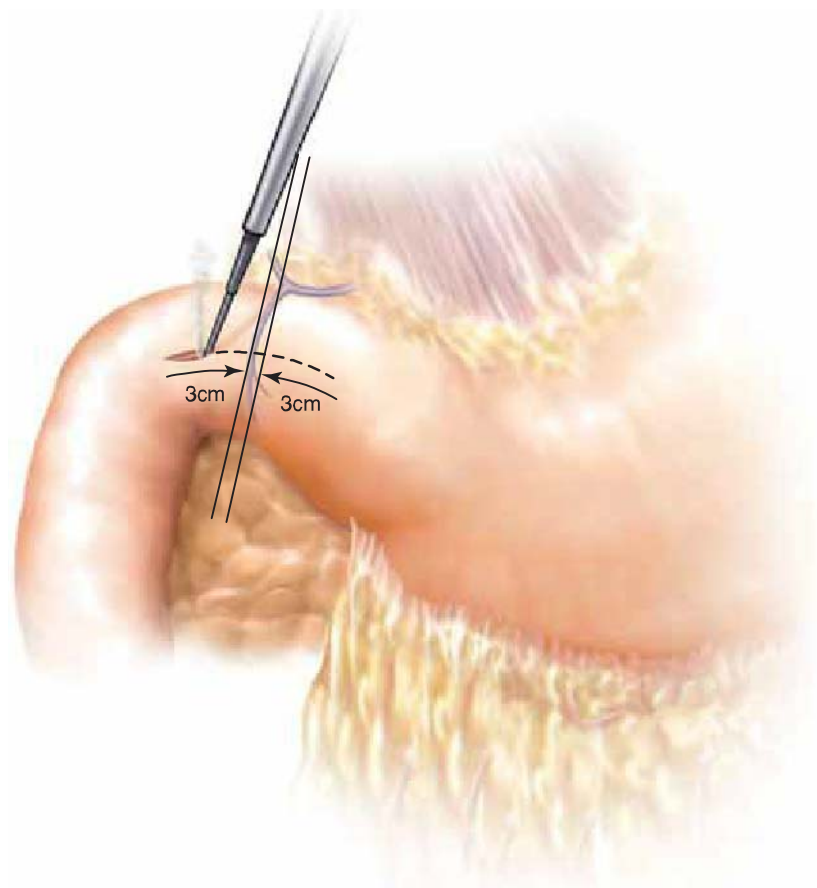


Figure 12.1 Traction sutures are placed using the “vein of Mayo” to help identify the pylorus. A 6-cm longitudinal is positioned midway at the pylorus to adequately expose the posterior duodenal bulb and identify the bleeding duodenal ulcer.

Figure 12.2 Tension on the traction sutures allows visualization of the posterior duodenal wall and the bleeding gastroduodenal artery. Adequate suture ligation requires three sutures: superior, inferior, and the “U-stitch” to control the lateral pancreatic branch.

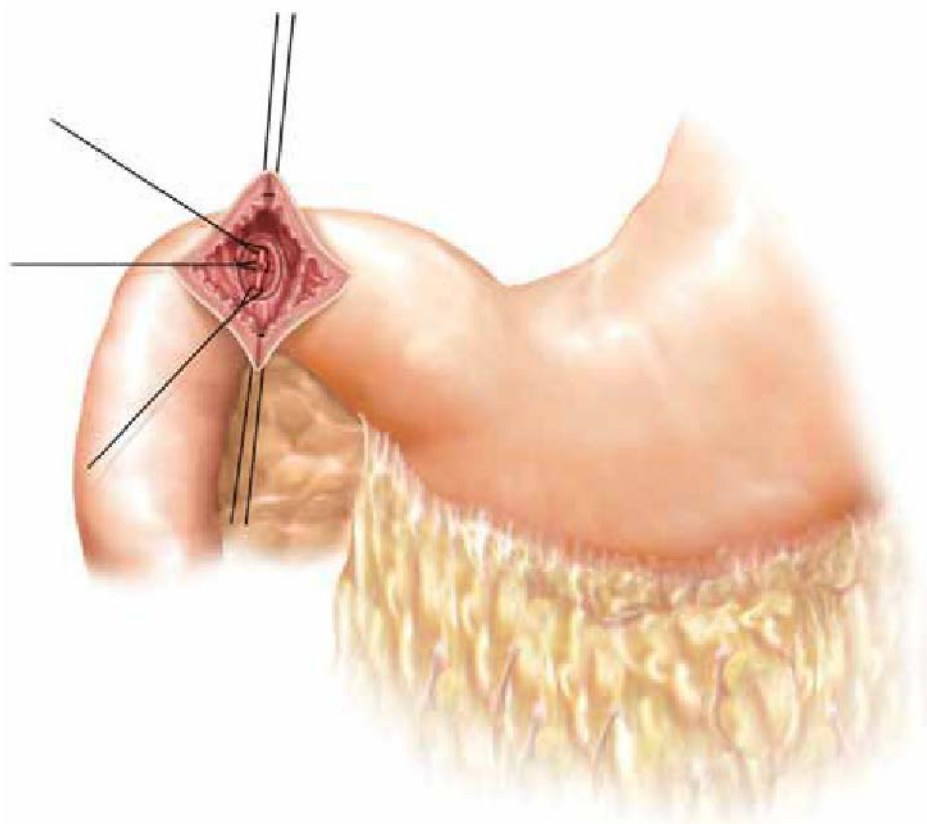
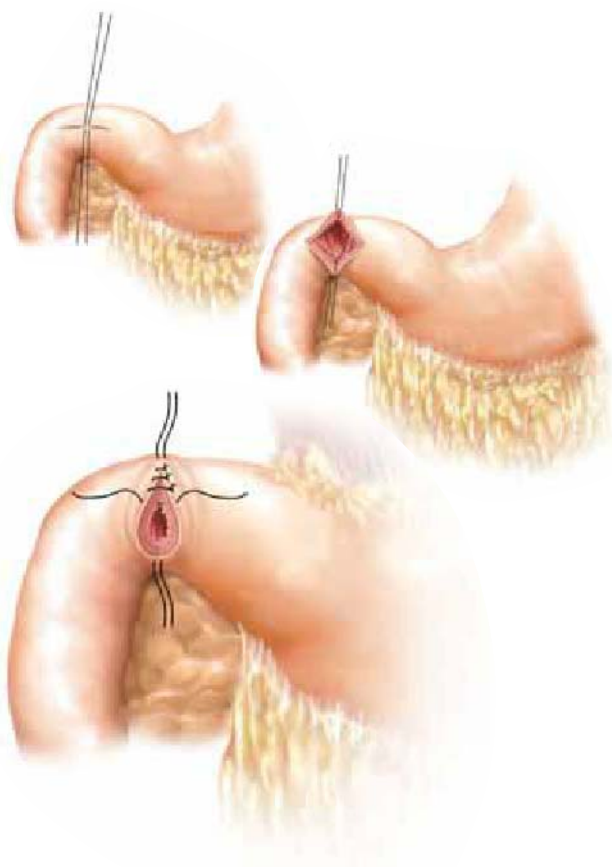


Figure 12.3 A Heineke–Mikulicz pyloroplasty is performed using the traction sutures as the superior and the inferior corners of the transverse closure.



closure with 3-0 silk sutures can be either single- or two-layered. With a single-layered closure, the corners usually appear “dog-eared,” tempting surgeons to add a second layer. Care must be taken to avoid narrowing the lumen if the corners are turned in or a second layer of Lembert sutures are placed. If there appears to be excessive tension on the closure, the duodenum should be mobilized by freeing the lateral attachments with a full Kocher maneuver. A tongue of omentum can be mobilized to buttress the repair.

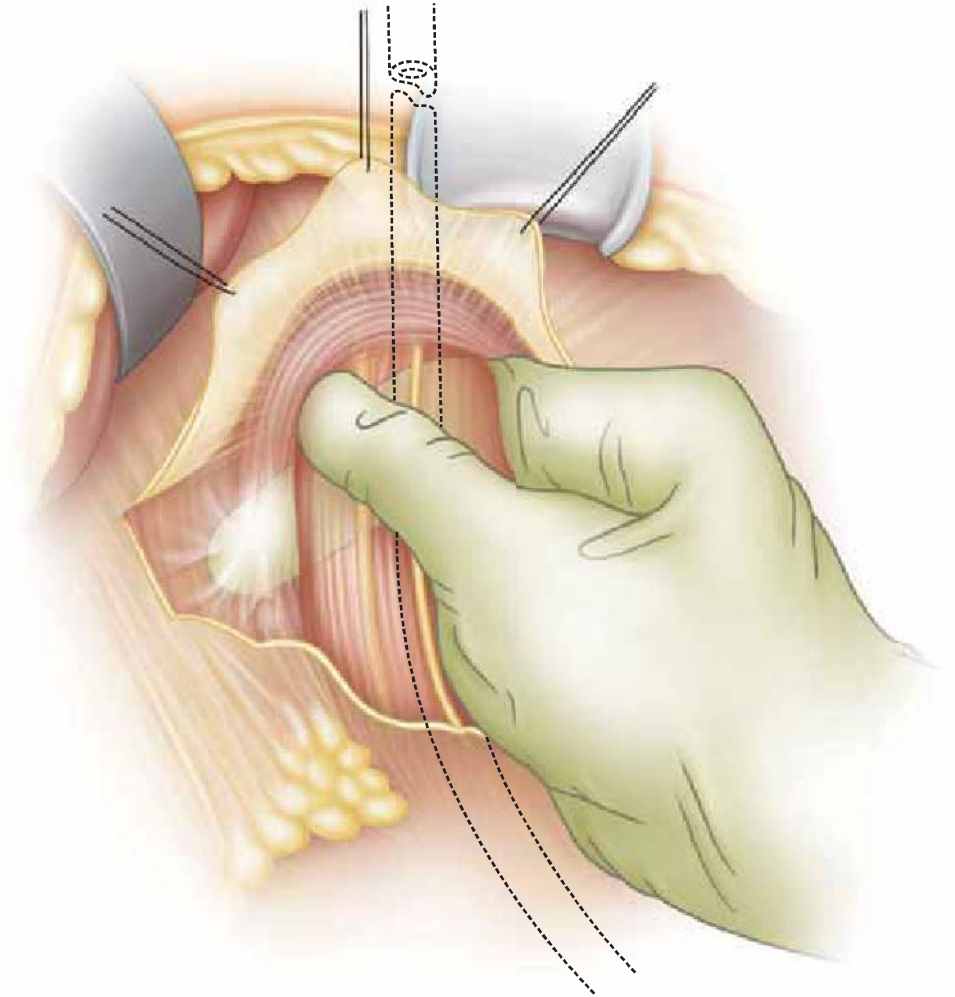
Truncal Vagotomy

The choice of vagotomy is highly dependent on the surgeon’s experience and the patient’s characteristics; however, truncal vagotomy is the procedure of choice for patients requiring surgical treatment for bleeding duodenal ulcer. The operating surgeon should stand on the patient’s right side. This procedure is started by gaining adequate exposure to the upper abdomen and esophageal hiatus. A liver retractor for the Bookwalter is used to elevate the left lobe of the liver. Another option is to free the triangular ligament and fold the left hepatic lobe medially. The stomach is retracted inferiorly with Babcock clamps. The pars flaccida is opened and proximally divided toward the phrenoesophageal ligament which is likewise divided transversely at the esophageal hiatus (Fig. 12.4). The anterior mediastinum is entered with gentle manual blunt dissection using the thumb and index finger to develop the mediastinal dissection (Fig. 12.5). The esophagus is circumferentially dissected using the NG tube as a guide, and the anterior and posterior vagus branches are identified. The placement of a Penrose drain around the esophagus is sometimes useful for inferior retraction of the esophagus. The anterior vagus branch is usually directly anterior at the hiatus and gently courses to the patient’s left as it ascends in the mediastinum. The posterior branch is usually toward the right posterior side and is the more difficult of the two branches to identify. If there is any question as to the location of this branch, additional posterior mediastinal dissection is needed to identify it. After identification, both nerve branches are sharply dissected away from the esophagus, and a 2-cm specimen is resected from both (Figs. 12.6 and 12.7). Intraoperative frozen section can be performed to verify nerve tissue if any doubt exists, otherwise permanent specimens suffice. Additional anterior and posterior branches are divided by skeletonizing the esophagus for at least 2 cm



Figure 12.4 Truncal vagotomy is begun by opening the phrenoesophageal membrane transversely.

Figure 12.5 Gentle manual dissection with the thumb and index finger is used to create the retro-esophageal window using the nasogastric tube as a guide to free the esophagus circumferentially.



circumferentially, ensuring that the posterior nerve of Grassi is taken. A final check of hemostasis in the mediastinum is made before the esophageal hiatus is approximated with interrupted permanent suture (Fig. 12.8). Placement of closed suction drains is not required. The midline fascia and skin are closed in standard fashion.



POSTOPERATIVE MANAGEMENT

Patients should be admitted to and monitored in a surgical intensive care unit. A NG tube should be left in the midbody of the stomach for 2 to 3 days for stomach decompression. Prolonged NG tube placement should be avoided as early refeeding is associated with improved ulcer healing time. The patient should be kept on intravenous proton pump inhibitors and treated appropriately for *H. pylori* infection if found. A contrast study may be obtained prior to NG removal.



COMPLICATIONS

Rebleeding can occur in up to 10% of patients and is usually a technical failure due to inadequate suture placement at the original surgery if it recurs early on. Late rebleeding is usually associated with inadequate vagotomy or failure to eradicate *H. pylori* infection.

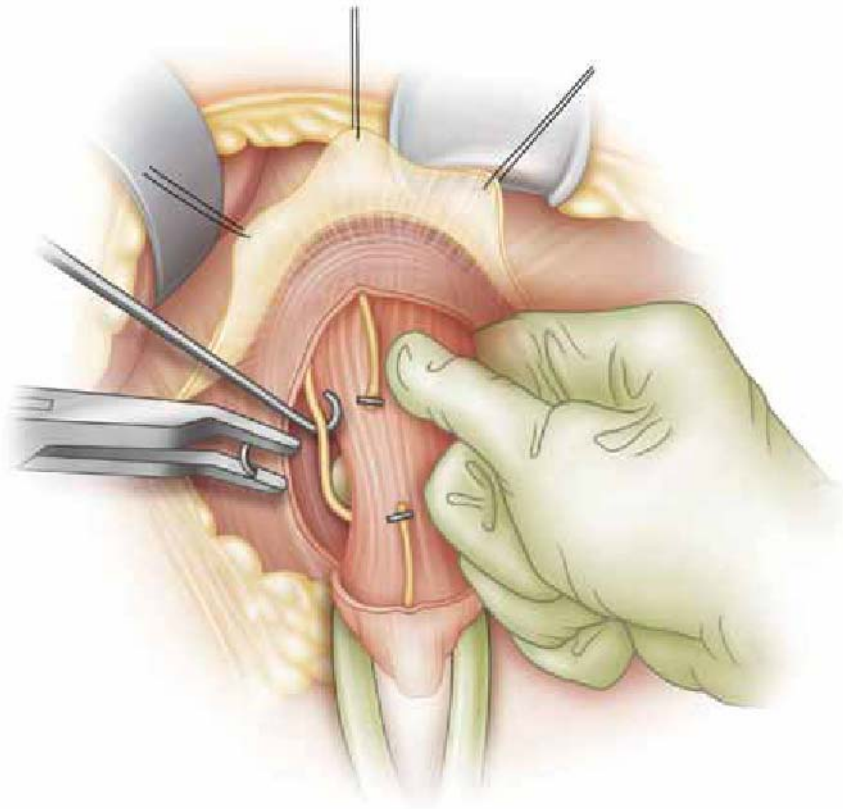


Figure 12.6 After the anterior vagus nerves is identified, a 2-cm segment is resected.

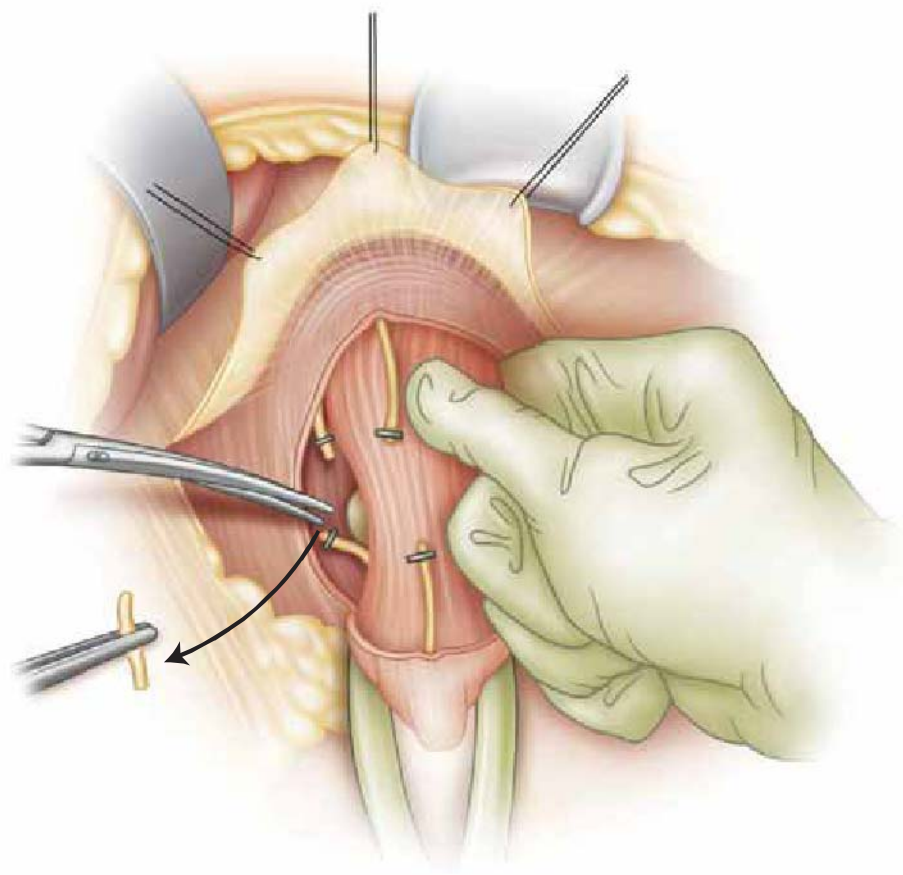
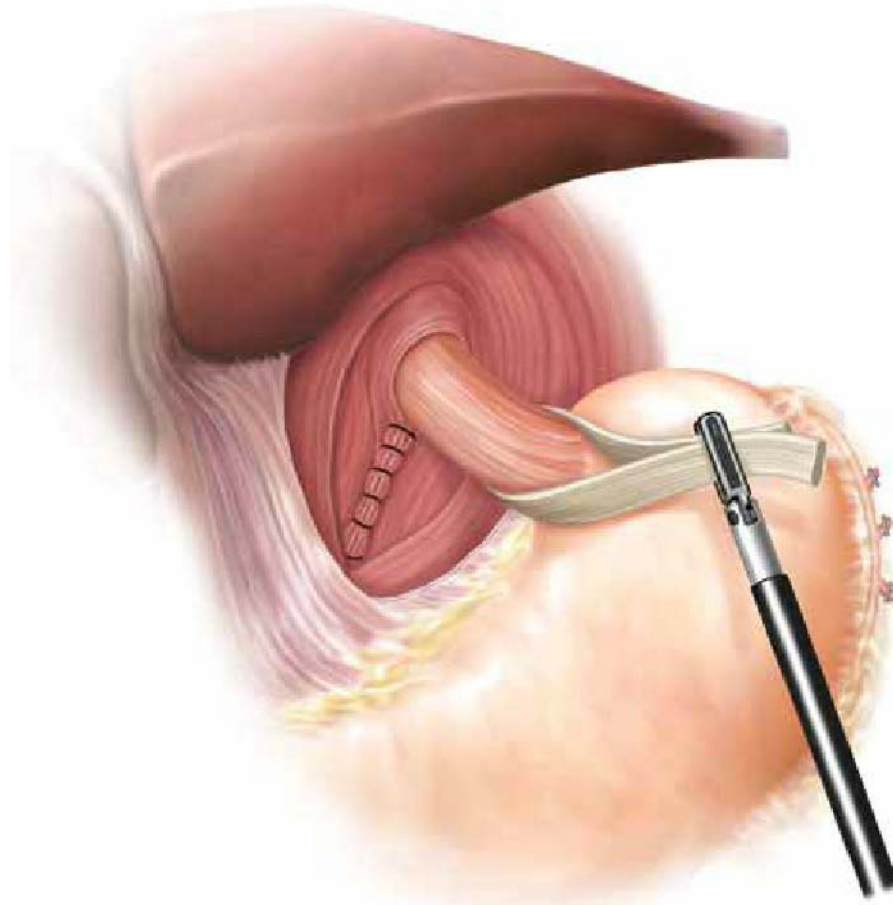


Figure 12.7 After the 2-cm segment of posterior vagus nerves is identified, the distal esophagus is then skeletonized to ensure a complete vagotomy.

Figure 12.8 The posterior crura are approximated with interrupted permanent 2-0 sutures allowing easy passage of an instrument alongside the esophagus.



Postvagotomy diarrhea can occur in up to 30% of patients following truncal vagotomy and may be related to the passage of unconjugated bile salts into the colon. Cholestyramine (which binds bile salts) can be used in patients to help control symptoms, but most cases are self-limited.

Dumping syndrome can occur in about 15% of patients after vagotomy and pyloroplasty and can manifest in a variety of symptoms such as bloating, cramping, diarrhea, and emesis. The etiology of this is thought to be due to the rapid emptying of hyperosmotic liquid into the small intestine and should be managed by dietary measures. Dumping symptoms are also frequently self-limited and improve with time.



CONCLUSIONS

Pyloroplasty and truncal vagotomy is required in the armamentarium of every general surgeon. It is straightforward, easily reproduced, and effective in the treatment of patients with bleeding duodenal ulcer.

Recommended References and Readings

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